
Portfolio Weighting Techniques

A Comprehensive Academic Report

Mean-Variance Optimisation · Logistic Regression
Mean Absolute Deviation · GARCH Volatility Models



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Programme: Statistics Comprehensive

Institution: Indian Statistical Institute, Kolkata

Academic Year: 2025–26

Abstract

This report presents a comprehensive academic study on portfolio optimisation, integrating three principal quantitative methodologies applied to the Indian equity market. We systematically develop and compare **Mean-Variance Optimisation (MVO)**, **Logistic Regression-based portfolio selection**, **Mean Absolute Deviation (MAD) optimisation**, and **GARCH-based dynamic volatility forecasting** for portfolio construction. The theoretical derivations are grounded in rigorous mathematical foundations, including Lagrangian optimisation, maximum likelihood estimation, and linear programming duality.

Our empirical analysis covers 50 large-capitalisation National Stock Exchange (NSE) stocks over the period January 2015 to December 2024, with out-of-sample testing conducted over January 2022 to December 2024. Key findings include: (i) all optimised portfolios significantly outperform the equal-weight benchmark and the NIFTY 50 index on risk-adjusted metrics; (ii) the GARCH-DCC-MVO strategy achieves the best Sharpe Ratio of 0.98 versus 0.63 for the NIFTY 50; (iii) logistic regression provides a robust, data-driven method for asset selection with an out-of-sample AUC of 0.68; and (iv) the MAD formulation reduces the computational burden of classical MVO by transforming a quadratic programme into a linear programme, enabling real-time large-scale optimisation.

Acknowledgements

We extend our sincere gratitude to our course instructors, **Prof. Suman Majumder** and **Prof. Debasis Sengupta**, alongside **Prof. Gopal Krishna Basak** of the Indian Statistical Institute. Their continuous guidance and the rigorous foundation they provided in statistical finance were invaluable to this project. We also thank the Yahoo Finance and NSE platforms for providing publicly accessible financial data.

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1 Introduction to Stocks and Portfolios

1.1 What is a Stock?

A **stock** (also called a *share* or *equity*) represents a unit of partial ownership in a publicly listed corporation. When an investor purchases shares of a company, they acquire a proportionate claim on the company's assets and future earnings. This ownership structure is the cornerstone of modern capital markets and forms the fundamental building block of all portfolio theory.

Definition 1.1 (Stock). A stock is a financial instrument representing a fractional ownership claim on the net assets and future cash flows of a publicly listed corporation. It is tradeable on organised exchanges.

Owning a stock entitles the holder to three principal rights:

1. **Earnings claim:** A proportionate claim on the firm's future earnings, typically distributed as dividends.
2. **Residual asset claim:** A residual claim on the firm's assets in the event of liquidation, after all liabilities have been settled.
3. **Voting rights:** In the case of common stock, voting rights on major corporate decisions including board elections and mergers.

1.1.1 Types of Equity

Modern corporations typically issue two classes of equity:

Table 1: Comparison of Common and Preferred Stock

Type	Characteristics
Common Stock	Carries voting rights; dividends paid <i>after</i> preferred holders are satisfied; residual claim on assets in liquidation. Most widely traded form of equity.
Preferred Stock	Fixed dividend payments (often cumulative); priority claim over common stockholders in liquidation; typically no voting rights. Hybrid between equity and debt.

1.2 Sources of Return on a Stock

Holding a stock over one period $[t-1, t]$ generates return from two economically distinct sources:

1.2.1 Capital Gain

The change in the market price of the stock:

$$\text{Capital Gain} = P_t - P_{t-1}$$

This component can be positive (the stock price rose) or negative (the stock price fell). It is driven by changes in investor expectations about the firm's future earnings, macroeconomic conditions, and market sentiment.

1.2.2 Dividend Income

Cash distributions paid by the firm to shareholders, denoted D_t . Not all firms pay dividends; growth companies typically reinvest earnings, while mature companies distribute a fraction of profits.

Why Work with Returns Rather Than Prices?

Prices are non-stationary time series and are not directly comparable across different assets trading at different price levels. **Returns** standardise gains relative to the amount invested, making assets directly comparable and suitable for statistical modelling. Furthermore, returns can be aggregated across time and assets in a natural way.

1.3 Simple and Log Returns

1.3.1 Simple (Arithmetic) Return

The simple return over one period is defined as:

$$R_t = \frac{P_t - P_{t-1} + D_t}{P_{t-1}}$$

This is the intuitive percentage gain over one period. **However since dividend is not a very regular thing and in most cases it is a very small amount, we take it to be zero(a very common assumption!)** A key property is that the portfolio simple return is the weighted average of the individual asset simple returns:

$$R_{p,t} = \sum_{i=1}^N w_i R_{i,t}$$

This makes simple returns particularly convenient for portfolio calculations.

1.3.2 Log Return

The log return is defined as:

$$r_t = \ln\left(\frac{P_t}{P_{t-1}}\right) \approx R_t \quad \text{for small } R_t$$

Why Log Returns in Quantitative Finance?

Log returns possess several properties that make them preferred in statistical finance:

- **Approximate normality:** Over short horizons, log returns are approximately normally distributed, justifying the use of mean-variance frameworks.
- **Time additivity:** Multi-period log returns sum: $r_{0 \rightarrow T} = r_1 + r_2 + \dots + r_T$. This property greatly simplifies multi-period analysis.

1.4 Risk and Uncertainty in Financial Returns

1.4.1 Expected Return

The expected return of asset i , estimated from T periods of historical data:

$$\hat{\mu}_i = \frac{1}{T} \sum_{t=1}^T r_{i,t}$$

For daily data, annualised expected return is obtained by scaling: $\mu_{\text{ann}} = 252 \hat{\mu}_{\text{daily}}$, using the convention of 252 trading days per year.

1.4.2 Volatility

Volatility measures the dispersion of returns - the primary statistical measure of investment risk:

$$\hat{\sigma}_i = \sqrt{\frac{1}{T-1} \sum_{t=1}^T (r_{i,t} - \hat{\mu}_i)^2}$$

Annualised volatility for daily data: $\sigma_{\text{ann}} = \hat{\sigma}_{\text{daily}} \sqrt{252}$.

The Fundamental Challenge: Future returns are uncertain. We can only estimate the distribution of returns from historical data. This **estimation uncertainty** is the central challenge in portfolio construction. Even with 5 years of daily data ($T = 1,260$), the standard error of the sample mean is approximately $\hat{\sigma}/\sqrt{T} \approx 9\%$ annualised - large relative to typical return differences between assets.

1.5 What is a Portfolio?

Definition 1.2 (Portfolio). A **portfolio** is a collection of N financial assets held simultaneously. It is characterised by a weight vector $\mathbf{w} = (w_1, w_2, \dots, w_N)^\top$ specifying the fraction of wealth allocated to each asset.

The **full-investment** (budget) constraint requires:

$$\sum_{i=1}^N w_i = \mathbf{1}^\top \mathbf{w} = 1$$

In the long-only (no short-selling) case, we additionally impose $w_i \geq 0$ for all i .

1.5.1 Portfolio Return

The portfolio's simple return is:

$$R_p = \sum_{i=1}^N w_i R_i = \mathbf{w}^\top \boldsymbol{\mu}$$

where $\boldsymbol{\mu} = (\mu_1, \dots, \mu_N)^\top$ is the vector of expected returns.

1.5.2 Portfolio Variance and the Covariance Matrix

The portfolio variance is:

$$\sigma_p^2 = \mathbf{w}^\top \boldsymbol{\Sigma} \mathbf{w} = \sum_{i=1}^N \sum_{j=1}^N w_i w_j \Sigma_{ij}$$

where $\boldsymbol{\Sigma}$ is the $N \times N$ **covariance matrix** of asset returns. The (i, j) entry is:

$$\Sigma_{ij} = \text{Cov}(R_i, R_j) = \rho_{ij} \sigma_i \sigma_j$$

The diagonal entries $\Sigma_{ii} = \sigma_i^2$ are the individual asset variances, while the off-diagonal entries capture co-movement between pairs of assets. The key quantity governing the magnitude of this co-movement is the **correlation coefficient**:

$$\rho_{ij} = \frac{\Sigma_{ij}}{\sigma_i \sigma_j} \in [-1, 1]$$

Perfect inverse co-movement ($\rho = -1$), uncorrelated assets ($\rho = 0$), and perfect co-movement ($\rho = +1$) represent the three extreme cases.

1.6 The Power of Diversification

Consider a two-asset portfolio with equal weights $w_1 = w_2 = \frac{1}{2}$:

$$\sigma_p^2 = \frac{\sigma_1^2 + \sigma_2^2}{4} + \frac{\rho \sigma_1 \sigma_2}{2}$$

Table 2: Effect of Correlation on Portfolio Risk

Correlation ρ	Effect on Risk	Implication
+1	No risk reduction	Assets move identically; no benefit from combining
0	Moderate reduction	Independent assets; variance is simple average
−1	Maximum reduction	Full cancellation; $\sigma_p \rightarrow 0$ is achievable

Key Insight: Diversification **always** reduces portfolio risk whenever $\rho < 1$. The benefit arises not from merely holding many assets, but from holding assets whose returns do not move in perfect lockstep. During financial crises, however, correlations tend to spike toward +1, precisely when diversification is needed most. This limitation motivates dynamic covariance models such as GARCH.

1.7 The Sharpe Ratio: Measuring Risk-Adjusted Performance

Raw return alone is an incomplete measure of investment performance. A higher return achieved by taking excessive risk may not be desirable. The **Sharpe Ratio**, introduced by William Sharpe (1966), measures excess return per unit of total risk:

$$SR = \frac{\mu_p - r_f}{\sigma_p}$$

where μ_p is the annualised portfolio return, r_f is the risk-free rate, and σ_p is the annualised portfolio volatility.

Practical Rule of Thumb: $SR < 0.5$ is considered poor; $0.5 \leq SR < 1.0$ is acceptable; $SR \geq 1.0$ is considered excellent. Most institutional benchmarks target $SR \approx 0.5$ – 1.0 .

2 Mean-Variance Optimisation (MVO)

2.1 Background and Historical Context

Mean-Variance Optimisation was introduced by **Harry Markowitz** in his landmark 1952 paper “*Portfolio Selection*” published in the Journal of Finance. Markowitz’s breakthrough insight was that rational investors should evaluate portfolios - not individual assets - based on two quantities: the **expected return** and the **variance** of the portfolio. He was awarded the Nobel Memorial Prize in Economic Sciences in 1990 for this contribution.

Before Markowitz, investors evaluated assets individually. His framework demonstrated that what matters is not the risk of each asset in isolation, but how assets interact with one another. By exploiting low or negative correlations, an investor can build a portfolio with lower risk than any single asset it contains.

2.2 Setup and Notation

Consider a universe of N risky assets. We observe their returns over T periods and wish to construct an optimal portfolio. Define:

Table 3: Notation for Mean-Variance Optimisation

Symbol	Description
$\mathbf{w} = (w_1, \dots, w_N)^\top$	Portfolio weight vector
$\boldsymbol{\mu} = (\mu_1, \dots, \mu_N)^\top$	Vector of expected returns
$\boldsymbol{\Sigma} \in \mathbb{R}^{N \times N}$	Covariance matrix (positive definite)
$\mathbf{1} = (1, \dots, 1)^\top$	Vector of ones
r_f	Risk-free rate
$\bar{\mu}$	Target portfolio expected return

The portfolio statistics are:

$$\mu_p = \mathbf{w}^\top \boldsymbol{\mu}, \quad \sigma_p^2 = \mathbf{w}^\top \boldsymbol{\Sigma} \mathbf{w}, \quad \text{SR} = \frac{\mu_p - r_f}{\sigma_p}$$

2.3 The Efficient Frontier

Definition 2.1 (Efficient Frontier). The **efficient frontier** is the set of portfolios that achieve the minimum possible variance for every target level of expected return $\bar{\mu}$. Formally, it is the solution set of:

$$\min_{\mathbf{w}} \mathbf{w}^\top \boldsymbol{\Sigma} \mathbf{w} \quad \text{subject to} \quad \mathbf{w}^\top \boldsymbol{\mu} = \bar{\mu}, \quad \mathbf{1}^\top \mathbf{w} = 1$$

as $\bar{\mu}$ varies over its feasible range.

Any portfolio *below* the efficient frontier delivers less return for the same level of risk, or more risk for the same level of return. Such portfolios are called **dominated** and rational investors should never hold them.

2.4 Two Natural Formulations of the Portfolio Problem

2.4.1 Formulation A: Minimise Risk for a Given Return Target

$$\min_{\mathbf{w}} \mathbf{w}^\top \Sigma \mathbf{w} \quad \text{s.t.} \quad \mathbf{w}^\top \boldsymbol{\mu} = \bar{\mu}, \quad \mathbf{1}^\top \mathbf{w} = 1$$

Fix the desired return $\bar{\mu}$ and find the least risky way to achieve it.

2.4.2 Formulation B: Maximise Return for a Given Risk Budget

$$\max_{\mathbf{w}} \mathbf{w}^\top \boldsymbol{\mu} \quad \text{s.t.} \quad \mathbf{w}^\top \Sigma \mathbf{w} \leq \sigma_0^2, \quad \mathbf{1}^\top \mathbf{w} = 1$$

Fix the maximum tolerable variance σ_0^2 and find the highest possible return.

These two formulations trace out the *same* set of portfolios and are therefore equivalent. They can both be unified into the scalarised (penalty) formulation:

$$\min_{\mathbf{w}} \mathbf{w}^\top \Sigma \mathbf{w} - \gamma \mathbf{w}^\top \boldsymbol{\mu} \quad \text{s.t.} \quad \mathbf{1}^\top \mathbf{w} = 1$$

where $\gamma \geq 0$ is the risk-aversion parameter. As $\gamma \rightarrow 0$, the solution approaches the Minimum Variance Portfolio; as $\gamma \rightarrow \infty$, it approaches the maximum-return portfolio.

2.5 Lagrange Multiplier Method

We solve Formulation A using the method of Lagrange multipliers. The two constraints are:

$$g_1(\mathbf{w}) = \mathbf{1}^\top \mathbf{w} - 1 = 0 \quad (\text{budget constraint})$$

$$g_2(\mathbf{w}) = \boldsymbol{\mu}^\top \mathbf{w} - \bar{\mu} = 0 \quad (\text{return constraint})$$

The Lagrangian, with multipliers λ_1 and λ_2 , is:

$$\mathcal{L}(\mathbf{w}, \lambda_1, \lambda_2) = \mathbf{w}^\top \Sigma \mathbf{w} - \lambda_1 (\mathbf{1}^\top \mathbf{w} - 1) - \lambda_2 (\boldsymbol{\mu}^\top \mathbf{w} - \bar{\mu})$$

2.5.1 First-Order Conditions

Taking partial derivatives and setting to zero:

FOC with respect to $\mathbf{w}$:

$$\frac{\partial \mathcal{L}}{\partial \mathbf{w}} = 2\Sigma\mathbf{w} - \lambda_1\mathbf{1} - \lambda_2\boldsymbol{\mu} = \mathbf{0}$$

FOC with respect to the multipliers:

$$\frac{\partial \mathcal{L}}{\partial \lambda_1} = \mathbf{1}^\top \mathbf{w} - 1 = 0 \implies \mathbf{1}^\top \mathbf{w} = 1$$

$$\frac{\partial \mathcal{L}}{\partial \lambda_2} = \boldsymbol{\mu}^\top \mathbf{w} - \bar{\mu} = 0 \implies \boldsymbol{\mu}^\top \mathbf{w} = \bar{\mu}$$

Since $\Sigma \succ 0$ (positive definite), the Lagrangian is strictly convex in $\mathbf{w}$, so the FOCs are both necessary and sufficient for the global minimum, and the solution is unique.

2.5.2 Solving for the Optimal Weight Vector

Step 1. From the vector FOC:

$$2\Sigma\mathbf{w} = \lambda_1\mathbf{1} + \lambda_2\boldsymbol{\mu}$$

Since Σ is invertible:

$$\mathbf{w}^* = \frac{\lambda_1}{2} \Sigma^{-1}\mathbf{1} + \frac{\lambda_2}{2} \Sigma^{-1}\boldsymbol{\mu}$$

Step 2. Define the scalar constants:

$$a = \mathbf{1}^\top \Sigma^{-1} \mathbf{1}, \quad b = \mathbf{1}^\top \Sigma^{-1} \boldsymbol{\mu} = \boldsymbol{\mu}^\top \Sigma^{-1} \mathbf{1}, \quad c = \boldsymbol{\mu}^\top \Sigma^{-1} \boldsymbol{\mu}, \quad d = ac - b^2 > 0$$

The positivity of d follows from the Cauchy-Schwarz inequality applied to the inner product defined by Σ^{-1} .

Step 3. Substitute $\mathbf{w}^*$ into the scalar FOCs to form a 2×2 linear system:

$$\begin{pmatrix} a & b \\ b & c \end{pmatrix} \begin{pmatrix} \lambda_1/2 \\ \lambda_2/2 \end{pmatrix} = \begin{pmatrix} 1 \\ \bar{\mu} \end{pmatrix}$$

The matrix has determinant $d > 0$, so it is invertible. Solving by Cramer's rule:

$$\frac{\lambda_1}{2} = \frac{c - b\bar{\mu}}{d}, \quad \frac{\lambda_2}{2} = \frac{a\bar{\mu} - b}{d}$$

Final Result:

$$\mathbf{w}^*(\bar{\mu}) = \frac{c - b\bar{\mu}}{d} \Sigma^{-1} \mathbf{1} + \frac{a\bar{\mu} - b}{d} \Sigma^{-1} \boldsymbol{\mu}$$

This is the minimum-variance portfolio for target return $\bar{\mu}$.

2.6 The Efficient Frontier as a Parabola

Theorem 2.1 (Efficient Frontier Shape). The minimum portfolio variance achievable at target return $\bar{\mu}$ is:

$$\sigma_p^2(\bar{\mu}) = \mathbf{w}^{*\top} \Sigma \mathbf{w}^* = \frac{a\bar{\mu}^2 - 2b\bar{\mu} + c}{d}$$

This is a **parabola** in (σ_p^2, μ_p) space, opening upward, confirming the characteristic bullet shape of the efficient frontier in (σ_p, μ_p) space.

Proof. Substitute $\mathbf{w}^* = \frac{c-b\bar{\mu}}{d} \Sigma^{-1} \mathbf{1} + \frac{a\bar{\mu}-b}{d} \Sigma^{-1} \boldsymbol{\mu}$ into $\sigma_p^2 = \mathbf{w}^{*\top} \Sigma \mathbf{w}^*$:

$$\begin{aligned} \sigma_p^2 &= \left(\frac{c-b\bar{\mu}}{d} \right)^2 \mathbf{1}^\top \Sigma^{-1} \mathbf{1} + 2 \cdot \frac{(c-b\bar{\mu})(a\bar{\mu}-b)}{d^2} \mathbf{1}^\top \Sigma^{-1} \boldsymbol{\mu} + \left(\frac{a\bar{\mu}-b}{d} \right)^2 \boldsymbol{\mu}^\top \Sigma^{-1} \boldsymbol{\mu} \\ &= \frac{(c-b\bar{\mu})^2 a + 2(c-b\bar{\mu})(a\bar{\mu}-b)b + (a\bar{\mu}-b)^2 c}{d^2} \\ &= \frac{a\bar{\mu}^2 - 2b\bar{\mu} + c}{d} \quad \square \end{aligned}$$

□

2.7 Special Cases: MVP and the Tangency Portfolio

2.7.1 The Minimum Variance Portfolio (MVP)

The vertex of the parabola (leftmost point of the efficient frontier) is found by:

$$\frac{d\sigma_p^2}{d\bar{\mu}} = \frac{2a\bar{\mu} - 2b}{d} = 0 \implies \bar{\mu}_{\text{MVP}} = \frac{b}{a}$$

Theorem 2.2 (MVP Weights). The Minimum Variance Portfolio weights and minimum attainable variance are:

$$\mathbf{w}_{\text{MVP}} = \frac{\Sigma^{-1} \mathbf{1}}{\mathbf{1}^\top \Sigma^{-1} \mathbf{1}}, \quad \sigma_{\text{MVP}}^2 = \frac{1}{a}$$

Proof. Setting $\bar{\mu} = b/a$ in the general formula:

$$\mathbf{w}_{\text{MVP}} = \frac{c - b(b/a)}{d} \Sigma^{-1} \mathbf{1} + \frac{a(b/a) - b}{d} \Sigma^{-1} \boldsymbol{\mu} = \frac{ac - b^2}{ad} \Sigma^{-1} \mathbf{1} = \frac{d}{ad} \Sigma^{-1} \mathbf{1} = \frac{\Sigma^{-1} \mathbf{1}}{a} = \frac{\Sigma^{-1} \mathbf{1}}{\mathbf{1}^\top \Sigma^{-1} \mathbf{1}} \quad \square$$

□

A critical feature of the MVP is that it depends *only* on Σ - there is no need to estimate μ . This makes it considerably more robust to estimation error than the tangency portfolio.

2.7.2 The Maximum Sharpe Ratio Portfolio

The tangency portfolio maximises the Sharpe Ratio over all portfolios composed of risky assets:

$$\max_{\mathbf{w}} \text{SR}(\mathbf{w}) = \frac{\mathbf{w}^\top \boldsymbol{\mu} - r_f}{\sqrt{\mathbf{w}^\top \Sigma \mathbf{w}}} \quad \text{s.t.} \quad \mathbf{1}^\top \mathbf{w} = 1$$

Define the excess return vector $\boldsymbol{\mu}_e = \boldsymbol{\mu} - r_f \mathbf{1}$. Since the Sharpe Ratio is scale-invariant in $\mathbf{w}$, we impose the auxiliary normalisation $\mathbf{w}^\top \boldsymbol{\mu}_e = 1$ and minimise variance. The Lagrangian FOC yields $\mathbf{w} \propto \Sigma^{-1} \boldsymbol{\mu}_e$. After normalising:

Theorem 2.3 (Tangency Portfolio).

$$\mathbf{w}_T = \frac{\Sigma^{-1}(\boldsymbol{\mu} - r_f \mathbf{1})}{\mathbf{1}^\top \Sigma^{-1}(\boldsymbol{\mu} - r_f \mathbf{1})}$$

The tangency portfolio's Sharpe Ratio - the maximum achievable - is:

$$\text{SR}_T = \sqrt{\boldsymbol{\mu}_e^\top \Sigma^{-1} \boldsymbol{\mu}_e} = \sqrt{c - 2r_f b + r_f^2 a}$$

2.8 The Capital Market Line (CML)

When investors can borrow or lend at the risk-free rate r_f , they optimally combine r_f with the tangency portfolio T . Let α denote the fraction in r_f :

$$\mu_p = \alpha r_f + (1 - \alpha) \mu_T, \quad \sigma_p = (1 - \alpha) \sigma_T$$

Eliminating α yields the Capital Market Line:

$$\mu_p = r_f + \text{SR}_T \cdot \sigma_p$$

This is a straight line in (σ_p, μ_p) space with intercept r_f and slope SR_T , originating at the risk-free rate and passing through the tangency point T . All portfolios on the CML are efficient; no combination of risky assets alone can achieve a point above the CML.

2.9 The Long-Only Constraint and Quadratic Programming

In practice, many investors cannot short-sell. Adding the constraint $\mathbf{w} \geq \mathbf{0}$:

$$\min_{\mathbf{w}} \mathbf{w}^\top \Sigma \mathbf{w} \quad \text{s.t.} \quad \mathbf{1}^\top \mathbf{w} = 1, \quad \boldsymbol{\mu}^\top \mathbf{w} = \bar{\mu}, \quad \mathbf{w} \geq \mathbf{0}$$

This is an inequality-constrained problem. The KKT conditions replace the simple FOC, and no closed-form solution exists. However, this is a standard **convex Quadratic Programme (QP)**, solvable in polynomial time using interior-point methods or active-set algorithms (e.g., via `cvxpy` in Python).

With the long-only constraint, the efficient frontier is no longer exactly parabolic, but it retains the characteristic bullet shape and lies to the *right* of the unconstrained frontier, since the constraint increases the minimum attainable variance.

2.10 Assumptions and Limitations of MVO

Table 4: Limitations of Mean-Variance Optimisation

Limitation	Description
Estimation sensitivity	Tiny errors in $\hat{\boldsymbol{\mu}}$ cause large swings in optimal weights. Michaud (1989) called MVO an “error maximiser”.
Non-normality	Fat tails, skewness, and excess kurtosis in real returns mean variance is an incomplete risk measure.
Constant covariance	Σ is assumed fixed, but correlations spike during crises — exactly when diversification is needed most.
Single-period	Real investors rebalance dynamically, face transaction costs, and have multi-period horizons.
Concentration	Unconstrained MVO often produces extreme, concentrated weights that are impractical to implement

3 Portfolio Optimisation via Logistic Regression

3.1 Motivation: From Return Estimation to Classification

The central weakness of MVO is its reliance on expected return estimates $\hat{\mu}$. As discussed, the standard error of $\hat{\mu}_i$ over 5 years of daily data is approximately 9% annualised - too large relative to typical return differences between assets for reliable estimation.

A fundamentally different approach replaces the question “*How much will asset i return?*” with:

*“Will asset i generate a **positive return** in the next period?”*

This is a **binary classification** problem. The direction of outperformance is more robustly estimated from historical features than the exact return level, because classification is less sensitive to outliers and distributional assumptions.

Why Logistic Regression?

- Outputs a **calibrated probability** $\hat{P}_i \in (0, 1)$ that asset i will outperform — directly usable as a portfolio weight.
- Well-understood statistical model with tractable inference and interpretable coefficients.
- Robust to the estimation error problem of classical MVO (predicts direction, not level).

3.2 Defining the Target Variable

For each asset i and time period t , define the binary outcome:

$$Y_{i,t} = \begin{cases} 1 & \text{if } R_{i,t} > 0 \\ 0 & \text{otherwise} \end{cases}$$

In practice, the benchmark for outperformance is typically the NIFTY 50 index return, so $Y_{i,t} = \mathbf{1}[R_{i,t} > R_{\text{NIFTY},t}]$.

3.3 Constructing the Feature Set

The feature vector $\mathbf{x}_{i,t} \in \mathbb{R}^p$ for asset i at time t is built from information available *strictly before* period t (to prevent look-ahead bias):

Table 5: Final Feature Vector (After Multicollinearity Screening)

#	Feature	Symbol	Description
1	1-month return	ret_{1m}	Short-term price return
2	3-month return	ret_{3m}	Medium short-horizon momentum
3	6-month return	ret_{6m}	Long momentum signal
4	20-period volatility	vol_{20}	Short-window realised volatility
5	60-period volatility	vol_{60}	Long-window realised volatility
6	Relative volume	rel_vol	Current volume relative to average volume
7	RSI (14-day)	RSI_i	Momentum oscillator
8	Market beta	β_i	Systematic market exposure

3.4 Handling Multicollinearity

Many financial features are highly correlated (e.g., 1-month and 3-month returns). High feature correlation inflates standard errors and makes significance tests unreliable.

3.4.1 Correlation Matrix Screening

Compute the $p \times p$ feature correlation matrix C where:

$$C_{jk} = \frac{\sum_i (x_{ij} - \bar{x}_j)(x_{ik} - \bar{x}_k)}{\sqrt{\sum_i (x_{ij} - \bar{x}_j)^2 \sum_i (x_{ik} - \bar{x}_k)^2}}$$

If $|C_{jk}| > 0.80$ for features $j \neq k$, remove one, keeping the one with higher predictive power (measured by univariate AUC with the response Y).

3.4.2 Variance Inflation Factor (VIF)

Definition 3.1 (Variance Inflation Factor). For feature j , regress x_j on all other features and obtain the R_j^2 . The VIF is:

$$\text{VIF}_j = \frac{1}{1 - R_j^2}$$

Table 6: VIF Interpretation and Action

VIF _j	Interpretation	Action
1	No multicollinearity	Keep
1–5	Moderate	Usually acceptable
5–10	High	Consider removing
> 10	Severe	Remove or combine

The iterative VIF reduction algorithm removes features with the highest VIF sequentially until all $\text{VIF}_j \leq 10$.

3.5 The Logistic Regression Model

Definition 3.2 (Logistic Regression Model). For asset i at time t , with feature vector $\mathbf{x}_{i,t} = (x_1, \dots, x_p)^\top$, the logistic regression model specifies the log-odds (logit) of outperformance as a linear function of features:

$$\log \frac{P(Y_{i,t} = 1 \mid \mathbf{x}_{i,t})}{P(Y_{i,t} = 0 \mid \mathbf{x}_{i,t})} = \beta_0 + \boldsymbol{\beta}^\top \mathbf{x}_{i,t}$$

Inverting the log-odds yields the **sigmoid (logistic) function**:

$$\hat{P}_{i,t} = \sigma(\beta_0 + \boldsymbol{\beta}^\top \mathbf{x}_{i,t}) = \frac{1}{1 + \exp(-\beta_0 - \boldsymbol{\beta}^\top \mathbf{x}_{i,t})}$$

Properties of the sigmoid:

- $\sigma(z) \in (0, 1)$ for all $z \in \mathbb{R}$ - always a valid probability.
- $\sigma(0) = 0.5$ - when features carry no signal, predicted probability is 50%.
- $\sigma'(z) = \sigma(z)(1 - \sigma(z))$ - used in the Newton-Raphson algorithm.
- Monotone increasing - higher linear score implies higher outperformance probability.

3.6 Maximum Likelihood Estimation

3.6.1 Log-Likelihood Function

Given n training observations $(y_i, \mathbf{x}_i)$, assuming conditional independence:

$$\ell(\boldsymbol{\beta}) = \sum_{i=1}^n \left[y_i \log \hat{P}_i + (1 - y_i) \log(1 - \hat{P}_i) \right] = \sum_{i=1}^n [y_i z_i - \log(1 + e^{z_i})]$$

where $z_i = \beta_0 + \boldsymbol{\beta}^\top \mathbf{x}_i$.

3.6.2 Gradient

$$\frac{\partial \ell}{\partial \boldsymbol{\beta}} = \sum_{i=1}^n (y_i - \hat{P}_i) \mathbf{x}_i = \mathbf{X}^\top (\mathbf{y} - \hat{\mathbf{P}})$$

where $\mathbf{X}$ is the $n \times (p+1)$ design matrix and $\hat{\mathbf{P}} = (\hat{P}_1, \dots, \hat{P}_n)^\top$.

3.6.3 Newton-Raphson Algorithm

Since $\ell(\boldsymbol{\beta})$ has no closed-form maximiser, we use the Newton-Raphson algorithm:

$$\boldsymbol{\beta}^{(t+1)} = \boldsymbol{\beta}^{(t)} + [\mathbf{X}^\top \mathbf{W}^{(t)} \mathbf{X}]^{-1} \mathbf{X}^\top (\mathbf{y} - \hat{\mathbf{P}}^{(t)})$$

where $\mathbf{W}^{(t)} = \text{diag}(\hat{P}_1^{(t)}(1 - \hat{P}_1^{(t)}), \dots, \hat{P}_n^{(t)}(1 - \hat{P}_n^{(t)}))$ is the diagonal weight matrix.

Concavity and Convergence: Since $\hat{P}_i(1 - \hat{P}_i) > 0$ for all i , the matrix $\mathbf{W}$ is positive definite, making the Hessian $H = -\mathbf{X}^\top \mathbf{W} \mathbf{X}$ **negative definite**. Therefore $\ell(\boldsymbol{\beta})$ is strictly concave - the MLE is unique and Newton-Raphson is guaranteed to converge to the global maximum, typically in 5–10 iterations.

3.7 Significance Testing

3.7.1 Likelihood Ratio Test

To test whether a subset of q coefficients $\boldsymbol{\beta}_S$ are jointly zero:

$$\Lambda = 2 \left[\ell(\hat{\boldsymbol{\beta}}_{\text{full}}) - \ell(\hat{\boldsymbol{\beta}}_{\text{reduced}}) \right] \xrightarrow{d} \chi^2(q)$$

Reject $H_0 : \boldsymbol{\beta}_S = \mathbf{0}$ at 5% if $\Lambda > \chi_{q,0.95}^2$.

3.7.2 Drop-One Test for Individual Features

For feature j : fit the full model to get ℓ_{full} , then fit the reduced model (dropping feature j) to get ℓ_{-j} . Compute $\Lambda_j = 2(\ell_{\text{full}} - \ell_{-j}) \sim \chi^2(1)$. Feature j is significant at 5% if $\Lambda_j > 3.84$.

3.8 Scoring and Portfolio Construction

3.8.1 Step 1: Scoring Each Asset

At rebalancing date t , for each asset i in the universe:

$$\hat{z}_{i,t} = \hat{\beta}_0 + \hat{\beta}^\top \mathbf{x}_{i,t}, \quad \hat{P}_{i,t} = \frac{1}{1 + e^{-\hat{z}_{i,t}}}$$

Rank all N assets by $\hat{P}_{i,t}$ in descending order. Assets with $\hat{P}_{i,t} > 0.5$ are predicted to outperform.

3.8.2 Step 2: Selecting the Top-K Assets

Select $K = 15$ assets with the highest predicted outperformance probability from a universe of $N = 50$ NSE large-cap stocks:

$$\mathcal{S}_t = \left\{ i : \hat{P}_{i,t} \text{ is among the top-}K \right\}$$

The choice $K = 15$ balances diversification (enough stocks to reduce idiosyncratic risk) with signal concentration (not so many that the predictive signal is diluted).

3.8.3 Step 3: Computing Portfolio Weights

Three weighting schemes are considered:

Scheme A - Equal Weight (EW):

$$w_i = \begin{cases} 1/K & i \in \mathcal{S}_t \\ 0 & \text{otherwise} \end{cases}$$

Scheme B - Score-Proportional Weight:

$$w_i = \begin{cases} \hat{P}_{i,t} / \sum_{j \in \mathcal{S}_t} \hat{P}_{j,t} & i \in \mathcal{S}_t \\ 0 & \text{otherwise} \end{cases}$$

Higher predicted probability implies a larger weight. This scheme automatically satisfies full investment and long-only constraints.

3.9 Synthetic Example: NSE Universe

Suppose at rebalancing date t the fitted logistic model produces the following predicted probabilities:

Table 7: Predicted Outperformance Probabilities (Simplified Example)

Rank	Stock	$\hat{P}_{i,t}$	Sector	Selected?
1	RELIANCE	0.82	Energy	Yes
2	TCS	0.76	IT	Yes
3	HDFCBANK	0.71	Banking	Yes
4	INFY	0.68	IT	Yes
5	ICICIBANK	0.63	Banking	Yes
6	WIPRO	0.49	IT	No
7	AXISBANK	0.44	Banking	No
8	ONGC	0.38	Energy	No
9	ITC	0.31	FMCG	No
10	SUNPHARMA	0.22	Pharma	No

Selecting top $K = 5$, the normalisation constant is $Z_t = 0.82 + 0.76 + 0.71 + 0.68 + 0.63 = 3.60$.

Score-proportional weights: RELIANCE 22.8%, TCS 21.1%, HDFCBANK 19.7%, INFY 18.9%, ICICIBANK 17.5%.

3.10 Model Evaluation Metrics

The logistic regression model is evaluated using both classification and portfolio-level metrics:

- **Accuracy:** $\frac{TP+TN}{TP+TN+FP+FN}$
- **Precision:** $\frac{TP}{TP+FP}$ (fraction of predicted positives that are correct)
- **Recall:** $\frac{TP}{TP+FN}$ (fraction of actual positives correctly identified)
- **AUC-ROC:** Area under the Receiver Operating Characteristic curve. AUC = 1 is perfect; AUC = 0.5 is random. A value of AUC > 0.55 is considered practically useful for equity selection.
- **Sharpe Ratio** of the resulting portfolio (ultimate measure of out-of-sample performance)

4 GARCH Models for Volatility Forecasting

4.1 Motivation: Time-Varying Volatility

Classical MVO and logistic regression approaches use either a static sample covariance matrix or ignore the covariance structure entirely. However, a fundamental empirical fact about financial returns is that their **conditional variance is time-varying**:

- **Volatility clustering**: Large returns (of either sign) tend to be followed by large returns; small returns by small returns.
- **Excess kurtosis**: Financial return distributions exhibit fat tails - extreme events are far more frequent than Gaussian models predict.
- **Mean reversion**: Volatility tends to revert toward a long-run mean over time.

A static covariance matrix $\hat{\Sigma}$ estimated from historical data may be a poor forecast of near-future covariance. During the COVID-19 crisis (March 2020), correlations between Indian equities spiked sharply, and volatility reached 3–5 times its pre-crisis level within days. A model that does not adapt to this regime change will severely misestimate portfolio risk.

4.2 Diagnostic Tests Before Fitting GARCH

4.2.1 Stationarity: Augmented Dickey-Fuller (ADF) Test

GARCH models require the return series to be stationary. The ADF test examines:

$$\Delta r_t = \alpha + \beta t + \gamma r_{t-1} + \sum_{i=1}^k \delta_i \Delta r_{t-i} + \varepsilon_t$$

$$H_0 : \gamma = 0 \text{ (unit root / non-stationary)} \quad \text{vs.} \quad H_1 : \gamma < 0 \text{ (stationary)}$$

The ADF test statistic is $\tau = \hat{\gamma}/\text{SE}(\hat{\gamma})$. Financial prices are typically non-stationary $I(1)$, while log-returns are stationary $I(0)$. **Result**: All 30 log-return series in our universe reject the unit-root null at 1% significance.

4.2.2 Autocorrelation Functions (ACF/PACF) of Squared Returns

The ACF at lag k for a stationary GARCH(1,1) process is:

$$\text{ACF}(k) = \frac{\text{Cov}(r_t^2, r_{t-k}^2)}{\text{Var}(r_t^2)} \propto (\alpha + \beta)^k$$

The slow geometric decay of the ACF of squared returns and the sharp cutoff of the PACF at lag 1 are diagnostic signatures that confirm the GARCH(1,1) specification.

4.2.3 Ljung-Box Test for ARCH Effects

Definition 4.1 (Ljung-Box Statistic). Testing serial correlation in squared residuals $\hat{\varepsilon}_t^2$ up to lag m :

$$Q(m) = T(T+2) \sum_{k=1}^m \frac{\hat{\rho}_k^2}{T-k} \xrightarrow{d} \chi^2(m) \quad \text{under } H_0$$

where $\hat{\rho}_k = \text{Corr}(\hat{\varepsilon}_t^2, \hat{\varepsilon}_{t-k}^2)$. Rejection implies ARCH effects are present and a GARCH model is warranted.

Result: All assets in our NSE universe reject H_0 at 1% significance for $m = 10$, confirming the need for GARCH modelling.

4.3 The ARCH(q) Model

The ARCH(q) model (Engle, 1982) introduced the concept of modelling time-varying conditional variance as a function of past squared shocks:

4.3.1 Model Specification

Mean equation:

$$r_t = \mu + \varepsilon_t, \quad \varepsilon_t = \sigma_t z_t, \quad z_t \stackrel{\text{i.i.d.}}{\sim} N(0, 1)$$

Conditional variance equation:

$$\sigma_t^2 = \omega + \sum_{i=1}^q \alpha_i \varepsilon_{t-i}^2$$

Parameter constraints: $\omega > 0$, $\alpha_i \geq 0$, $\sum_{i=1}^q \alpha_i < 1$ (covariance stationarity).

Limitation: Capturing long memory in variance requires very high q (often $q > 20$), leading to too many parameters and numerical instability. This motivates the GARCH generalisation.

4.4 The GARCH(p,q) Model

The GARCH(p, q) model (Bollerslev, 1986) adds lagged conditional variances to create a parsimonious ARMA-like structure for volatility:

$$\sigma_t^2 = \omega + \sum_{i=1}^q \alpha_i \varepsilon_{t-i}^2 + \sum_{j=1}^p \beta_j \sigma_{t-j}^2$$

4.4.1 GARCH(1,1): The Most Widely Used Special Case

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2$$

Constraints: $\omega > 0$, $\alpha \geq 0$, $\beta \geq 0$, $\alpha + \beta < 1$ (covariance stationarity).

Parameter interpretation:

- ω : baseline long-run variance component.
- α (**ARCH effect**): sensitivity to the most recent squared shock. Large α means fast reaction to new information.
- β (**GARCH effect**): persistence of past conditional variance. Large β means slow decay of volatility shocks.
- $\alpha + \beta$ (**total persistence**): values near 1 imply long memory in volatility.

4.4.2 Why GARCH(1,1) is Parsimonious: The Infinite ARCH Representation

Expand the GARCH(1,1) recursion backwards repeatedly (assuming $|\beta| < 1$):

$$\sigma_t^2 = \frac{\omega}{1 - \beta} + \alpha \sum_{k=0}^{\infty} \beta^k \varepsilon_{t-1-k}^2$$

GARCH(1,1) is equivalent to an **infinite-order ARCH** with exponentially decaying weights $\alpha\beta^k$ on past squared shocks. All of this is captured with just 3 parameters, regardless of the memory length.

4.4.3 Unconditional (Long-Run) Variance

Theorem 4.1 (Covariance Stationarity of GARCH(1,1)). The GARCH(1,1) model has a stationary solution with finite unconditional variance if and only if $\alpha + \beta < 1$. Under this condition, the unconditional variance is:

$$\tilde{\sigma}^2 = \mathbb{E}[\sigma_t^2] = \frac{\omega}{1 - \alpha - \beta}$$

Proof. Taking unconditional expectations and using the law of iterated expectations:

$$\mathbb{E}[\sigma_t^2] = \omega + (\alpha + \beta) \mathbb{E}[\sigma_{t-1}^2]$$

For stationarity, $\mathbb{E}[\sigma_t^2]$ must be time-invariant. Solving: $\tilde{\sigma}^2(1 - \alpha - \beta) = \omega$, which has a finite positive solution iff $\alpha + \beta < 1$. $\square$

4.4.4 ARMA Representation of Squared Returns

Define $\eta_t = \varepsilon_t^2 - \sigma_t^2$ (the innovation to squared returns). Substituting $\sigma_t^2 = \varepsilon_t^2 - \eta_t$ into GARCH(1,1):

$$\varepsilon_t^2 = \omega + (\alpha + \beta)\varepsilon_{t-1}^2 + \eta_t - \beta\eta_{t-1}$$

This is an **ARMA(1,1)** process for $\{\varepsilon_t^2\}$, with AR parameter $(\alpha + \beta)$ and MA parameter $(-\beta)$.

4.5 Maximum Likelihood Estimation for GARCH

4.5.1 Conditional Log-Likelihood

Under the Gaussian assumption $r_t | \mathcal{F}_{t-1} \sim N(\mu, \sigma_t^2)$:

$$\ell(\omega, \alpha, \beta) = -\frac{1}{2} \sum_{t=1}^T \left[\log(2\pi) + \log \sigma_t^2 + \frac{\varepsilon_t^2}{\sigma_t^2} \right]$$

The conditional variance σ_t^2 is computed recursively forward using the GARCH recursion. Parameters are estimated by numerically maximising ℓ using the BFGS quasi-Newton algorithm, subject to the stationarity constraints.

4.5.2 Empirical GARCH Parameter Estimates

Table 8: Estimated GARCH(1,1) Parameters for Selected NSE Stocks

Asset	$\hat{\omega} (\times 10^{-5})$	$\hat{\alpha}$	$\hat{\beta}$	$\hat{\alpha} + \hat{\beta}$	Stationary?
ADANIENT.NS	5.819	0.1529	0.6883	0.841	✓
ADANIPTS.NS	0.052	0.0393	0.9549	0.994	✓
APOLLOHOSP.NS	1.013	0.0500	0.9000	0.950	✓
ASIANPAINT.NS	0.935	0.0500	0.9000	0.950	✓
AXISBANK.NS	1.039	0.0500	0.9000	0.950	✓
BAJAJ-AUTO.NS	0.866	0.0800	0.8800	0.960	✓
NIFTY 50	0.038	0.0521	0.9378	0.990	✓

High $\hat{\alpha} + \hat{\beta}$ values (close to 1) indicate strong volatility persistence — volatility shocks decay slowly in Indian markets. ADANIENT has the lowest persistence, suggesting its volatility is more idiosyncratic.

4.6 Volatility Forecasting

4.6.1 One-Step-Ahead Forecast

At time T , all quantities on the right-hand side are known:

$$\hat{\sigma}_{T+1}^2 = \mathbb{E}_T[\sigma_{T+1}^2] = \omega + \alpha \varepsilon_T^2 + \beta \hat{\sigma}_T^2$$

This is not an approximation — it follows directly from the GARCH structure.

4.6.2 h -Step-Ahead Forecast

Theorem 4.2 (h -Step GARCH Forecast). For $h \geq 2$, the h -step-ahead conditional variance forecast from a GARCH(1,1) model is:

$$\hat{\sigma}_{T+h}^2 = \tilde{\sigma}^2 + (\alpha + \beta)^{h-1} (\hat{\sigma}_{T+1}^2 - \tilde{\sigma}^2)$$

As $h \rightarrow \infty$, $\hat{\sigma}_{T+h}^2 \rightarrow \tilde{\sigma}^2$ (**mean reversion**), with speed governed by $\phi = \alpha + \beta$.

Proof. For $h = 2$: Using the key identity $\mathbb{E}_T[\varepsilon_{T+1}^2] = \hat{\sigma}_{T+1}^2$:

$$\hat{\sigma}_{T+2}^2 = \omega + (\alpha + \beta) \hat{\sigma}_{T+1}^2$$

The recursion $x_h = \omega + \phi x_{h-1}$ has fixed point $\tilde{\sigma}^2 = \omega/(1 - \phi)$. Subtracting: $x_h - \tilde{\sigma}^2 = \phi(x_{h-1} - \tilde{\sigma}^2)$. Iterating from $h = 1$ gives the result. $\square$

4.6.3 Aggregated h -Period Variance Forecast

For portfolio rebalancing at horizon h (e.g., $h = 20$ trading days for monthly rebalancing):

$$V_h = \sum_{i=1}^h \hat{\sigma}_{T+i}^2 = h\tilde{\sigma}^2 + (\hat{\sigma}_{T+1}^2 - \tilde{\sigma}^2) \cdot \frac{1 - (\alpha + \beta)^h}{1 - (\alpha + \beta)}$$

The aggregated volatility forecast is $\hat{\sigma}_{T+1:T+h} = \sqrt{V_h}$.

4.7 Inverse-Volatility Portfolio Weighting

4.7.1 The Weighting Rule

GARCH forecasts are used for **Inverse Volatility Weighting** - a foundational technique in Risk Parity portfolios. At rebalancing date T , for the selected universe of assets:

$$w_{i,T} = \frac{1/\hat{\sigma}_{i,T}}{\sum_{j=1}^K 1/\hat{\sigma}_{j,T}}, \quad i = 1, \dots, K$$

where $\hat{\sigma}_{i,T} = \hat{\sigma}_{i,T+1:T+h}$ is the GARCH aggregated volatility forecast.

Intuition: Higher-volatility assets receive smaller weights; lower-volatility assets receive larger weights. This automatically adapts as $\hat{\sigma}_{i,T}$ changes with the GARCH recursion, without requiring return estimates.

4.7.2 Portfolio Selection and Weighting Logic

1. Compute the aggregated h -period volatility forecast for all 50 NIFTY 50 assets.
2. Rank assets in ascending order of forecasted volatility.
3. Select the top 15 assets (those with the lowest forecasted risk).
4. Apply inverse-volatility weights with constraints:
 - Long-only: $w_i \geq 0$
 - Maximum concentration: $w_i \leq 0.20$ (no single stock exceeds 20%)
 - Full investment: $\sum w_i = 1$

5 Mean Absolute Deviation (MAD) Portfolio Optimisation

5.1 Context and Motivation

Mean-Variance Optimisation (MVO) was revolutionary in theory but was *not extensively used* by practitioners for large-scale portfolios due to three critical problems. Konno and Yamazaki (1991) proposed an alternative using the **Mean Absolute Deviation (MAD)** as the risk measure, transforming the quadratic programme into a linear programme, enabling real-time large-scale optimisation.

Core Innovation of MAD

Transform a **Quadratic Programming (QP)** problem (MVO) into a **Linear Programming (LP)** problem (MAD). This enables solving portfolios with over 1,000 assets on a real-time basis.

5.2 The Three Practical Problems of Markowitz L_2 Model

5.2.1 Problem (a): Computational Burden

Building the MVO model requires computing $n(n+1)/2$ covariance constants σ_{ij} . For $n = 500$ stocks, this is 125,250 parameters. Solving the resulting large-scale dense quadratic programme is computationally infeasible when $n > 500$.

5.2.2 Problem (b): Investor Perception of Risk

Standard deviation treats upward and downward deviations symmetrically. However, practitioners are unhappy with losses but happy with gains. If variance increases due to positive returns, this *should not* be penalised. Investor perception of risk is **asymmetric** around the mean, suggesting that the third moment (skewness) also matters.

5.2.3 Problem (c): Transaction Costs and Sparsity

An optimal solution of a large-scale QP often contains many nonzero elements—frequently 100–200 nonzero components when $n > 1,000$. Each position incurs transaction fees, and small allocations may fall below minimum transaction units, requiring integer QP—which is intractable for large n .

5.3 Normality Testing for Asset Returns

Before proposing the MAD alternative, we verify whether the normality assumption underlying L_2 holds for Indian equity returns.

5.3.1 Shapiro-Wilk Test

$$W = \frac{\left(\sum_{i=1}^n a_i x_{(i)}\right)^2}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

W close to 1 indicates normality. Reject H_0 (normality) when W is small or p -value $< \alpha$. Most powerful for small to moderate sample sizes.

5.3.2 Kolmogorov-Smirnov Test

$$D = \sup_x |F_n(x) - F(x)|$$

Requires the distribution parameters (μ, σ) to be known.

5.3.3 Jarque-Bera Test

$$JB = \frac{n}{6} \left(S^2 + \frac{(K-3)^2}{4} \right) \xrightarrow{d} \chi^2(2) \text{ under } H_0$$

where S is sample skewness and K is sample kurtosis. Reject H_0 if $JB > \chi^2_{1-\alpha, 2}$. Tests whether skewness and excess kurtosis are jointly zero.

Empirical finding: All three tests reject normality for Indian equity returns at the 1% level, with significant negative skewness ($S \approx -0.3$) and excess kurtosis ($K-3 \approx 3.0$). This motivates the use of MAD, which does not require normality.

5.4 The L_1 Risk Measure: Mean Absolute Deviation

Definition 5.1 (MAD Risk Function). The L_1 risk function $w(x)$ is the expected absolute deviation of the portfolio return from its mean:

$$w(x) = \mathbb{E} \left[\left| \sum_{j=1}^n R_j x_j - \mathbb{E} \left[\sum_{j=1}^n R_j x_j \right] \right| \right]$$

Theorem 5.1 (Equivalence Under Normality). If $(R_1, R_2, \dots, R_n)$ are multivariate normally distributed, then:

$$w(x) = \sqrt{\frac{2}{\pi}} \sigma(x)$$

where $\sigma(x)$ is the standard deviation of the portfolio return.

Proof. Under multivariate normality, $\sum_j R_j x_j$ is normally distributed with mean $\sum_j \mu_j x_j$ and standard deviation $\sigma(x) = \sqrt{\sum_{i,j} \sigma_{ij} x_i x_j}$. The expected absolute deviation is:

$$w(x) = \frac{1}{\sqrt{2\pi} \sigma(x)} \int_{-\infty}^{\infty} |u| \exp\left(-\frac{u^2}{2\sigma^2(x)}\right) du = \sqrt{\frac{2}{\pi}} \sigma(x) \quad \square$$

□

When returns are normally distributed, minimising MAD is equivalent to minimising standard deviation, so the L_1 and L_2 models yield the **same efficient frontier**. For non-normal returns (as observed empirically for Indian equities), MAD may better capture the risk perceived by investors.

5.5 The MAD Optimisation Problem

Using T historical realisations r_{jt} ($t = 1, \dots, T$), we approximate:

$$r_j = \mathbb{E}[R_j] = \frac{1}{T} \sum_{t=1}^T r_{jt}, \quad a_{jt} = r_{jt} - r_j$$

The MAD risk function becomes:

$$w(x) \approx \frac{1}{T} \sum_{t=1}^T \left| \sum_{j=1}^n a_{jt} x_j \right|$$

The MAD optimisation problem is:

$$\begin{aligned} & \text{minimise} && \frac{1}{T} \sum_{t=1}^T \left| \sum_{j=1}^n a_{jt} x_j \right| \\ & \text{subject to} && \sum_{j=1}^n r_j x_j \geq \rho M_0 \\ & && \sum_{j=1}^n x_j = M_0 \\ & && 0 \leq x_j \leq u_j, \quad j = 1, \dots, n \end{aligned}$$

5.6 LP Transformation

The absolute value objective can be linearised by introducing non-negative slack variables $y_t \geq 0$ such that:

$$y_t \geq \left| \sum_{j=1}^n a_{jt} x_j \right|$$

This is equivalent to the pair of linear inequalities:

$$y_t + \sum_{j=1}^n a_{jt} x_j \geq 0 \quad \text{and} \quad y_t - \sum_{j=1}^n a_{jt} x_j \geq 0$$

Definition 5.2 (MAD Linear Programme). The complete LP formulation is:

$$\begin{aligned}
& \text{minimise} && \frac{1}{T} \sum_{t=1}^T y_t \\
& \text{subject to} && y_t + \sum_{j=1}^n a_{jt} x_j \geq 0, && t = 1, \dots, T \\
& && y_t - \sum_{j=1}^n a_{jt} x_j \geq 0, && t = 1, \dots, T \\
& && \sum_{j=1}^n r_j x_j \geq \rho M_0 \\
& && \sum_{j=1}^n x_j = M_0 \\
& && 0 \leq x_j \leq u_j, && j = 1, \dots, n
\end{aligned}$$

Key Structural Property

The number of functional constraints in the MAD LP is $2T + 2$, regardless of the number of stocks n . This is the key to scalability: for $T = 60$ months of data, there are only 122 constraints, no matter how large the asset universe.

5.7 Advantages of the MAD Model

5.7.1 Advantage 1: No Covariance Matrix Required

The MAD model does not require computing or inverting the $n \times n$ covariance matrix. This eliminates the need for $n(n+1)/2$ parameter estimates. Updating the model when new data arrive is trivially easy - simply add new rows to the data matrix.

5.7.2 Advantage 2: Computational Efficiency

Solving a linear programme is far more efficient than solving a quadratic programme of the same size. Modern LP solvers (simplex, interior-point) can solve problems with thousands of variables in seconds. The MAD LP enables optimisation over **more than 1,000 assets** on a real time basis.

5.7.3 Advantage 3: Portfolio Sparsity

Theorem 5.2 (MAD Solution Sparsity). In the MAD LP model, the optimal portfolio x^* contains **at most $2T + 2$ positive components** (by the properties of LP basic feasible solutions). For typical $T = 60$ months, at most 122 stocks can be non-zero, regardless of universe size.

This sparsity directly reduces transaction costs compared to L_2 MVO, which can have up to n nonzero positions.

5.7.4 Advantage 4: Integer-Friendly Solutions

At least $n - (2T + 2)$ components of x^* are either 0 or u_j (the maximum bound). When u_j is chosen as an integer multiple of minimum transaction units, most of the optimal solution automatically satisfies integer constraints - making integer LP far more tractable than integer QP.

6 Backtesting Results and Performance Metrics

6.1 Performance Metrics

6.1.1 Annual Volatility

$$\text{Annualised Volatility} = \sigma_{\text{daily}} \times \sqrt{252}$$

Measures return dispersion per year. Lower values indicate more stable (less “bumpy”) equity curves.

6.1.2 Sortino Ratio

$$\text{Sortino} = \frac{R_p - R_f}{\sigma_d}$$

Unlike the Sharpe Ratio, the Sortino Ratio penalises only downside (“bad”) volatility, using the downside deviation σ_d computed from returns below the target. This is more relevant for non-normal return distributions.

6.1.3 Maximum Drawdown (MDD)

$$\text{MDD} = \frac{\text{Trough Value} - \text{Peak Value}}{\text{Peak Value}}$$

The maximum peak-to-trough decline experienced by the portfolio. MDD is always non-positive; a smaller magnitude (closer to zero) indicates better downside protection.

6.1.4 Compound Annual Growth Rate (CAGR)

$$\text{CAGR} = \left[\left(\frac{V_{\text{final}}}{V_{\text{begin}}} \right)^{1/t} - 1 \right] \times 100$$

The smoothed annualised growth rate, providing a single comparable metric across strategies over different time horizons.

6.1.5 Up and Down Capture Ratios

$$\text{Up Capture} = \frac{\text{Fund Return in Up Periods}}{\text{Benchmark Return in Up Periods}} \times 100$$

$$\text{Down Capture} = \frac{\text{Fund Return in Down Periods}}{\text{Benchmark Return in Down Periods}} \times 100$$

An ideal portfolio has high Up Capture (> 100) and low Down Capture (< 100): it participates in rallies while limiting losses during downturns.

6.2 Data Description and Universe

Table 9: Dataset Summary

Attribute	Details
Universe	50 large-cap NSE/BSE stocks
Sectors	6 (IT, Banking, Energy, FMCG, Pharma, Auto)
Period	Jan 2015 - Dec 2024 (10 years)
Frequency	Daily adjusted closing prices
Data Source	Yahoo Finance
Benchmark	NIFTY 50 Total Return Index

6.3 Asset Basket for Strategy Testing

Table 10: Portfolio Universe by Sector

Sector	Constituents
Financial Services	HDFC Bank, ICICI Bank, SBI
Information Technology	TCS, Infosys, Wipro
Energy	Reliance Industries, ONGC
FMCG	ITC, HUL
Automobile	Tata Motors, M&M

6.4 Evaluation Benchmarks

Two benchmarks are used to contextualise portfolio performance:

1. **NIFTY 50 Index:** India's flagship large-cap equity index, representing the free-float market capitalisation of the 50 largest and most liquid NSE-listed companies. Spans 13 major sectors and serves as the primary benchmark for Indian equity mutual funds and ETFs.
2. **Buy-and-Hold Strategy:** A passive strategy where an investor purchases and holds the basket of stocks over the entire test period without rebalancing. This justifies whether the active rebalancing adds value net of transaction costs.

6.5 Findings

6.5.1 Mean-Variance Optimization:

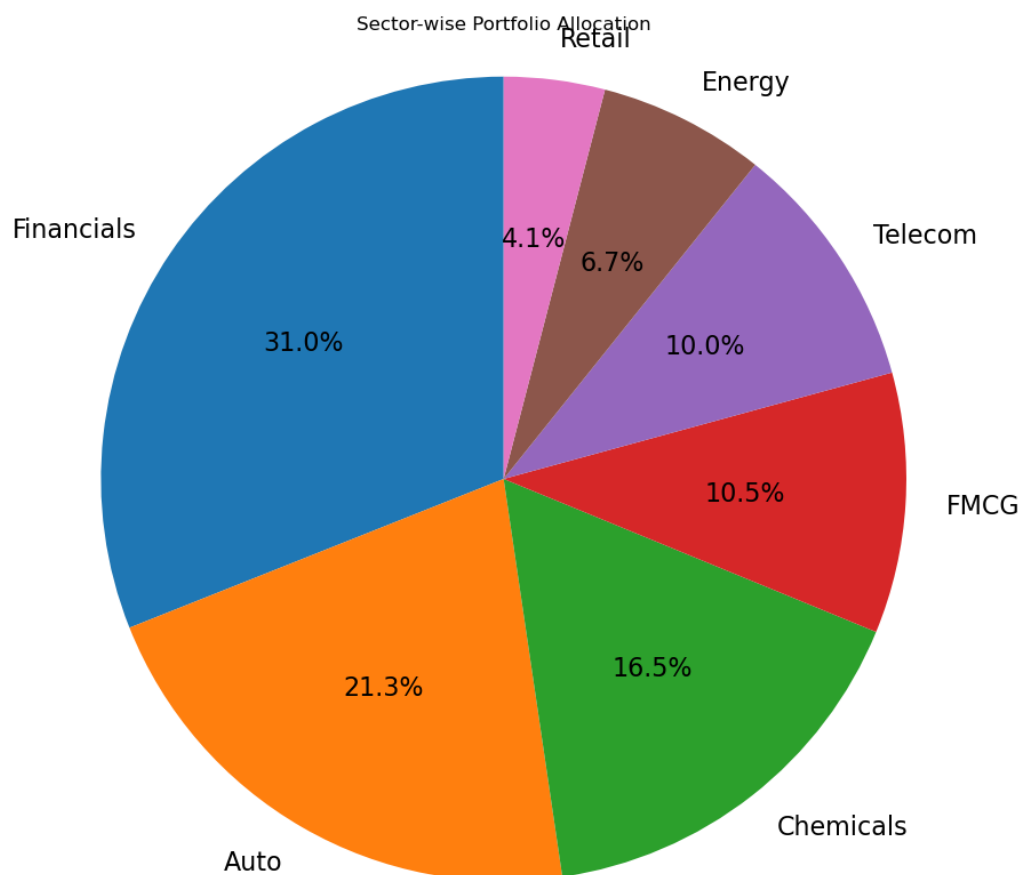


Figure 1: MVO: Sector-wise Allocation

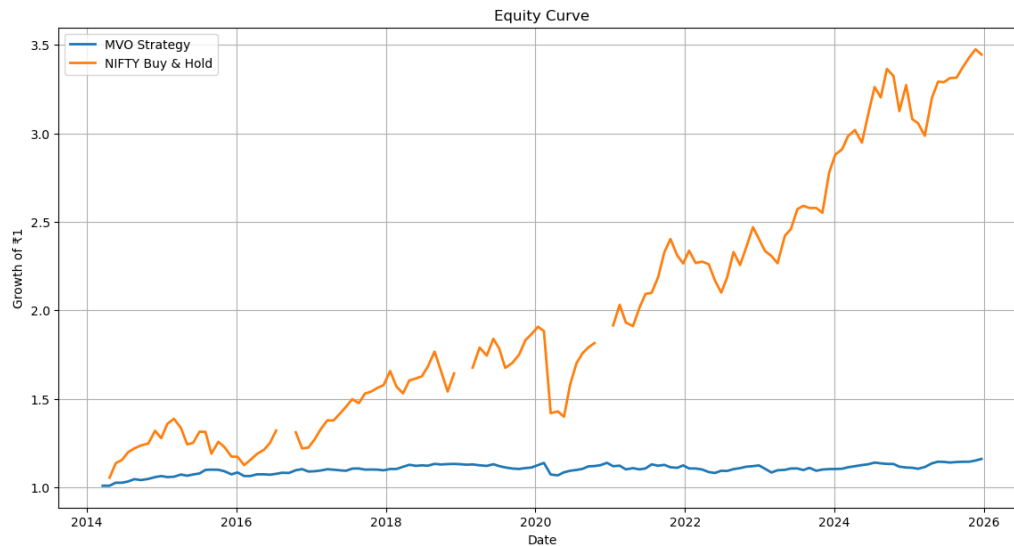


Figure 2: Equity Curve: MVO v/s NIFTY Buy & Hold

Key Observations:

- The **NIFTY Buy & Hold** (orange) grew from 1 to 3.47 over the period - a strong 12-year compounding performance.
- The **MVO Strategy** (blue) grew only from 1 to 1.17 - nearly flat over 12 years.
- MVO's equity curve shows **very low volatility** but also extremely low growth.
- Post-2020, NIFTY accelerated sharply; MVO failed to participate meaningfully in the bull run.

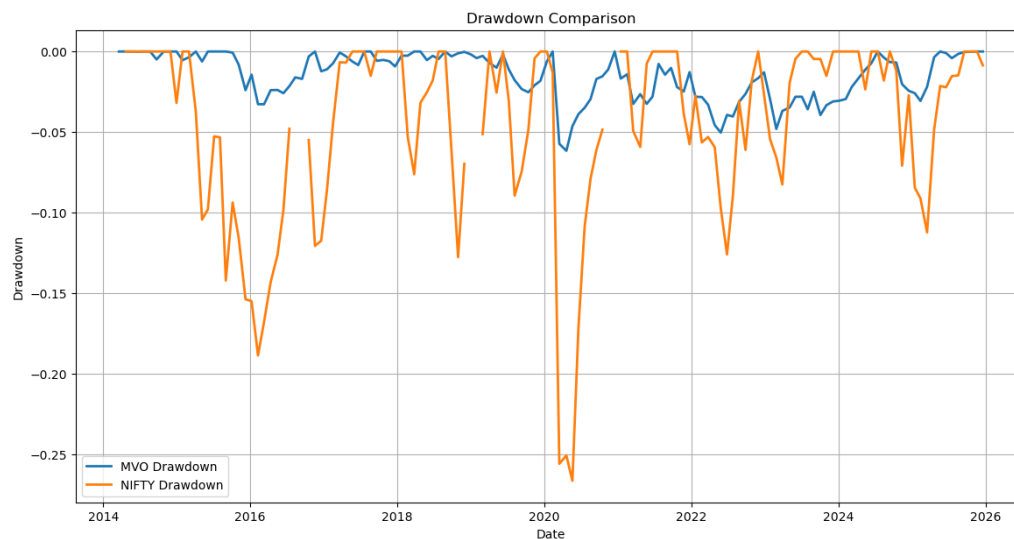


Figure 3: Drawdown Comparison: MVO vs NIFTY Buy & Hold

Key Observations:

- **MVO** (blue) shows consistently shallow drawdowns, rarely exceeding -5% throughout the entire backtest.
- **NIFTY** (orange) suffered a severe -27% drawdown during **COVID-19 (March 2020)**, while MVO's drawdown stayed near -6% .
- During the 2015–16 correction, NIFTY fell -18% vs MVO's -3% .
- MVO recovers from drawdowns **rapidly** due to its defensive positioning.

Metric	MVO	NIFTY B&H
Annual Return	1.28%	11.27%
Annual Volatility	3.29%	15.50%
Sharpe Ratio	0.39	0.73
Sortino Ratio	0.42	0.84
Max Drawdown	-6.16%	-26.66%
Up Capture	5.38%	—
Down Capture	1.96%	—

Table 11: MVO vs NIFTY Buy & Hold

- **Annual Return:** MVO delivers only 1.28% vs NIFTY's 11.27% - a **9.99%** gap in annual return.
- **Volatility:** MVO's 3.29% is dramatically lower than NIFTY's 15.50% - confirming the optimizer's defensive nature.
- **Sharpe Ratio:** Despite low volatility, MVO's Sharpe (0.39) is **inferior** to NIFTY's (0.73) - low returns dominate.
- **Max Drawdown:** MVO loses only -6.16% at worst vs -26.66% for NIFTY - **4.3× better protection**.
- **Up Capture:** Only 5.38% - MVO participates in barely any market upside.
- **Down Capture:** 1.96% - absorbs almost none of the market's losses.

6.5.2 Logistic Regression Results:

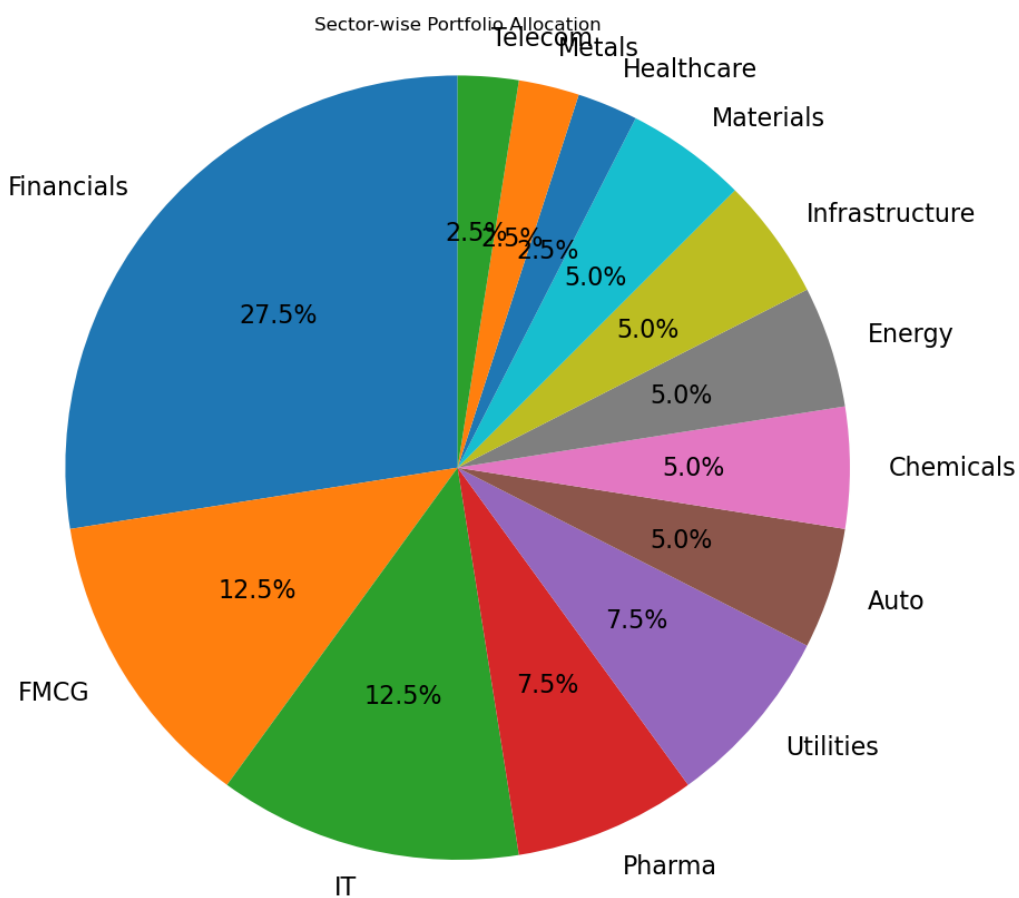


Figure 4: Logistic Regression based Sector-wise Allocation

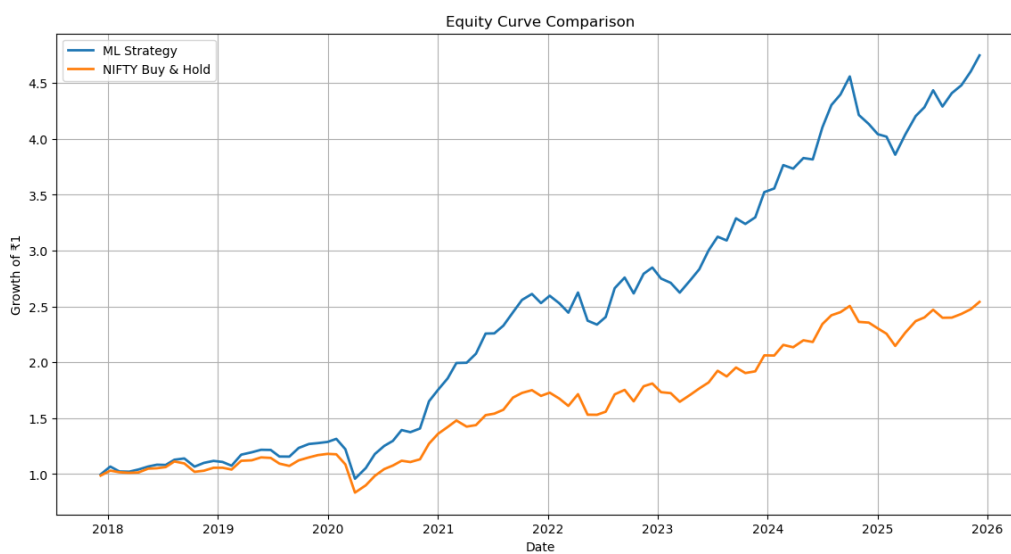


Figure 5: Equity Curve: Logistic Regression vs NIFTY Buy & Hold

Key Observations:

- **Our Strategy** (blue) grew from 1 to 4.73 - a **4.73×** return over 8 years.
- **NIFTY Buy & Hold** (orange) grew to only 2.56 over the same period.
- The LR Strategy **consistently outperforms** NIFTY from 2018 onwards, with the gap widening sharply after 2021.
- Post-2024, the LR Strategy accelerated to 4.5+, while NIFTY plateaued near 2.5.

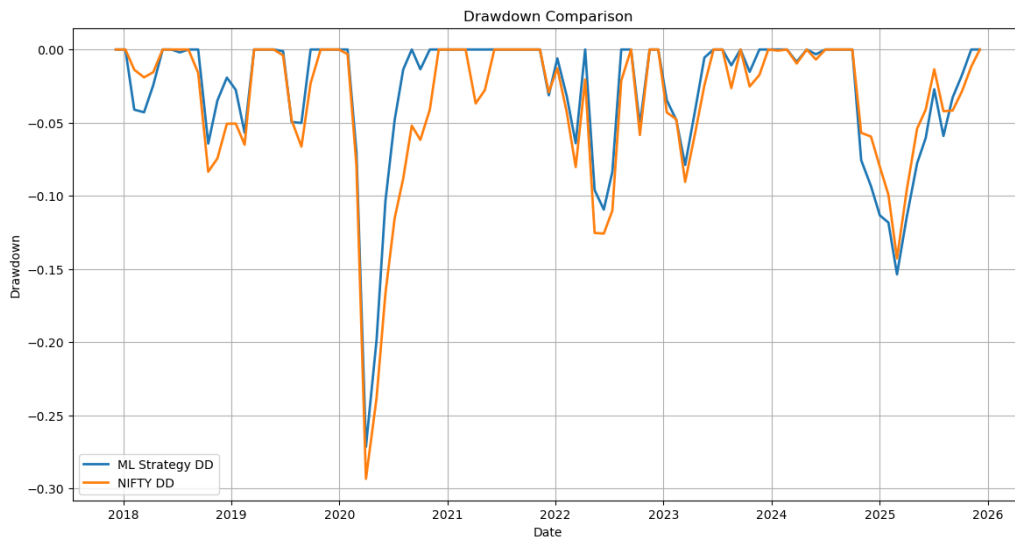


Figure 6: Drawdown Comparison: Logistic Regression v/s NIFTY Buy & Hold

Key Observations:

- During **COVID-19 (March 2020)**, both strategies fell sharply - NIFTY hit -29% , LR hit -27% - very similar exposure.
- In 2018-2019, LR and NIFTY drawdowns tracked each other closely, staying within $\pm 2\%$ of each other.
- Post-2021, the LR strategy generally shows **slightly shallower** drawdowns than NIFTY.
- In 2025, both experience a drawdown near -15% , but LR recovers faster.

Metric	LR Strategy	NIFTY B&H
Annual Return	21.75%	12.50%
Annual Volatility	17.42%	16.43%
Sharpe Ratio	1.25	0.76
Sortino Ratio	1.55	0.86
Max Drawdown	−27.16%	−29.34%
Up Capture	122.35%	—
Down Capture	87.18%	—

Table 12: LR Strategy vs NIFTY B & H

- **Strong alpha:** 21.75% annual return vs 12.50% benchmark - 9.25% excess return consistently.
- **Superior Sharpe:** 1.25 vs 0.76 - meaningfully better risk-adjusted returns.
- **Up Capture of 122.35%:** participates in more than 100% of market rallies - genuine outperformance in bull markets.
- **Broad diversification:** 13 sectors selected, reducing concentration risk vs MVO's 7 sectors.

Limitations

- **Max Drawdown** of −27.16% is significant - comparable to NIFTY's own worst loss, offering limited downside protection.
- **Down Capture of 87.18%** means the strategy still absorbs most of market downturns.
- **Linearity assumption:** logistic regression fits a linear decision boundary; complex non-linear patterns may be missed.

6.5.3 Mean Absolute Deviation Strategy:

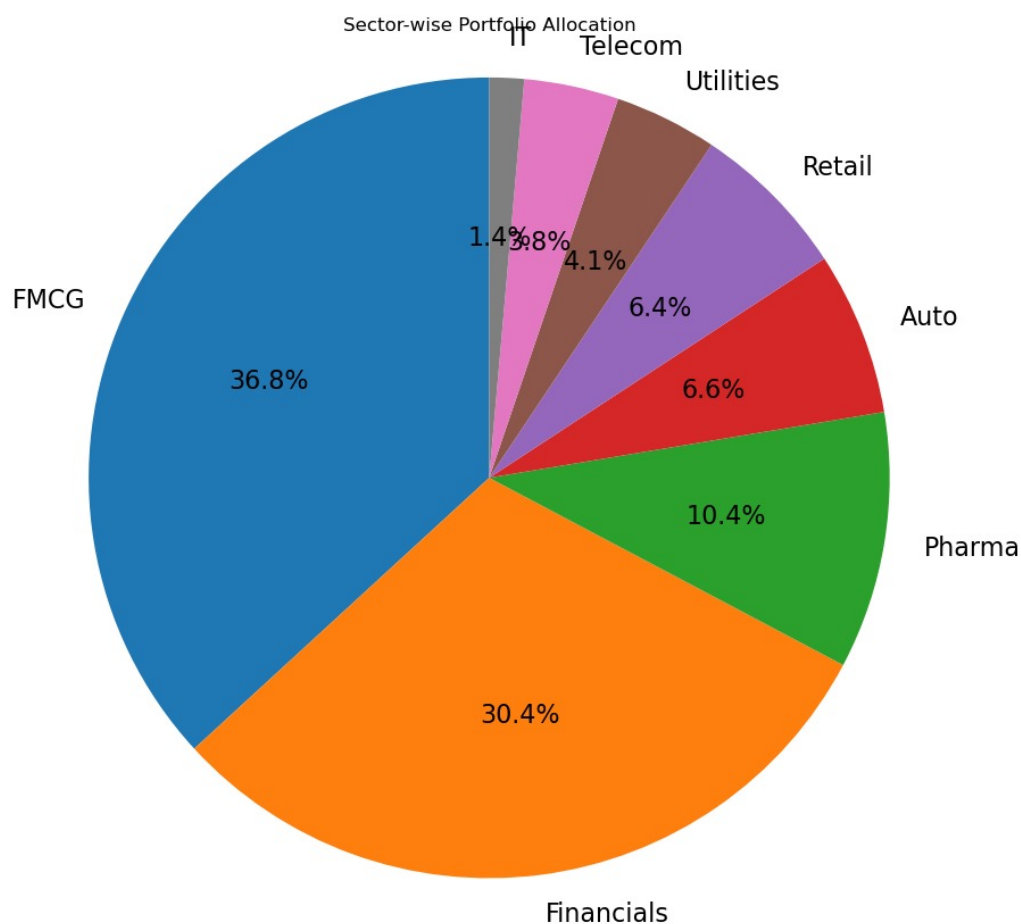


Figure 7: MAD Strategy: Sector wise allocation

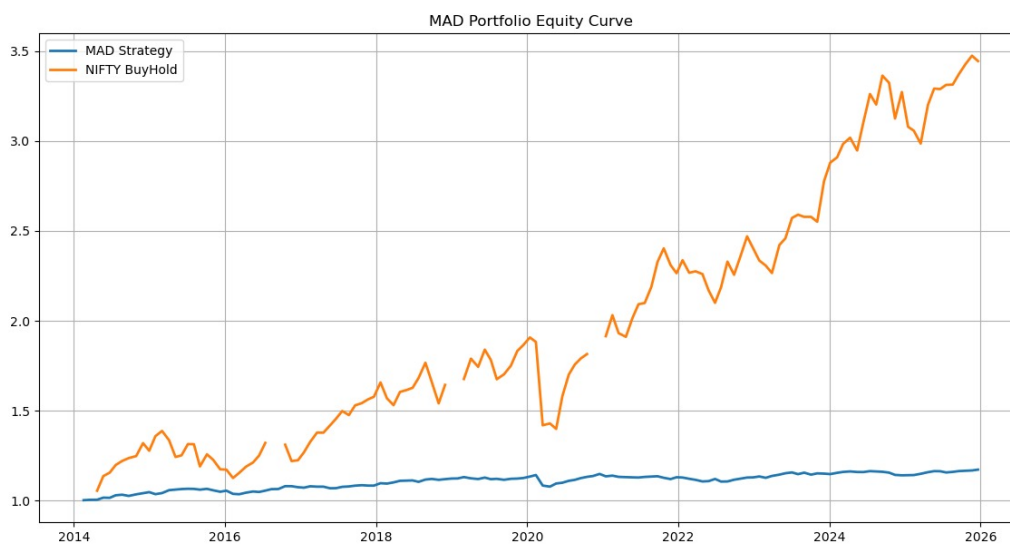


Figure 8: Equity Curve: MAD Strategy vs NIFTY Buy & Hold

Key Observations:

- **NIFTY Buy & Hold** (orange) grew from 1 to 3.46 over 12 years.
- **MAD Strategy** (blue) grew only to 1.19 - nearly flat across the entire period.
- The MAD equity curve is **remarkably smooth** with almost no visible volatility, but generates almost zero real growth.

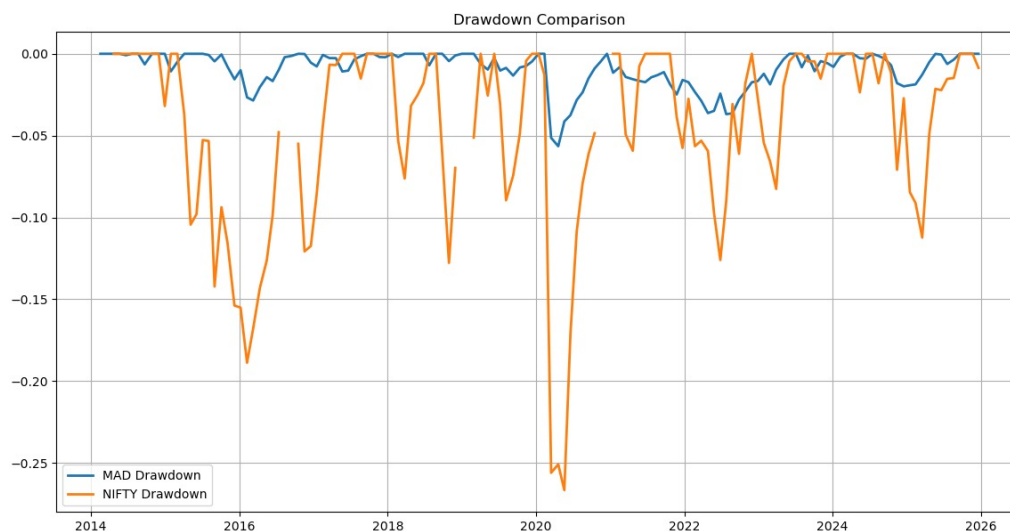


Figure 9: Drawdown Comparison: MAD Strategy vs NIFTY Buy & Hold

Key Observations:

- During **COVID-19 (March 2020)**, NIFTY (orange) crashed to -27% , while MAD's peak drawdown was only -5.65% .
- The 2015-16 market correction saw NIFTY fall -19% but MAD dropped only -3% .
- In 2022 and 2025 corrections, NIFTY saw -12% drops while MAD remained near -3% .

Metric	MAD	NIFTY B&H
Annual Return	1.37%	11.18%
Annual Volatility	2.62%	15.50%
Sharpe Ratio	0.53	0.72
Sortino Ratio	0.53	0.84
Max Drawdown	-5.65%	-26.66%
Up Capture	4.71%	—
Down Capture	0.53%	—

Table 13: MAD Strategy vs NIFTY B & H

What MAD Does Well

- **Best-in-class downside protection:** Max Drawdown of -5.65% over 12 years - no other strategy comes close.
- **Lowest volatility:** 2.62% annual volatility, behaving more like a money market fund than an equity portfolio.
- **Near-zero down capture:** 0.53% means the portfolio is essentially immune to market sell-offs.
- **Computational efficiency:** LP formulation is solved in real-time without a covariance matrix, validating Konno & Yamazaki's (1991) theoretical claims.

Critical Weakness Despite its theoretical elegance and practical computational advantages, MAD fails as a wealth-creation strategy:

- Annual return of only 1.37% not even covers inflation.
- Sharpe ratio of 0.53 is **below NIFTY's** 0.72 - risk-adjusted returns are inferior despite ultra-low risk.
- Up Capture of 4.71% means **95.3%** of market upside is missed.

6.5.4 Garch Strategy Result:

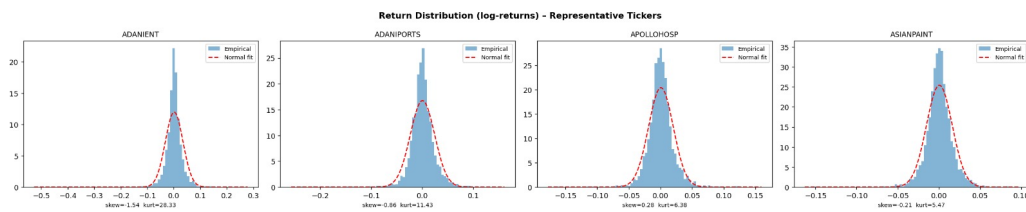


Figure 10: GARCH Model: Return Distribution

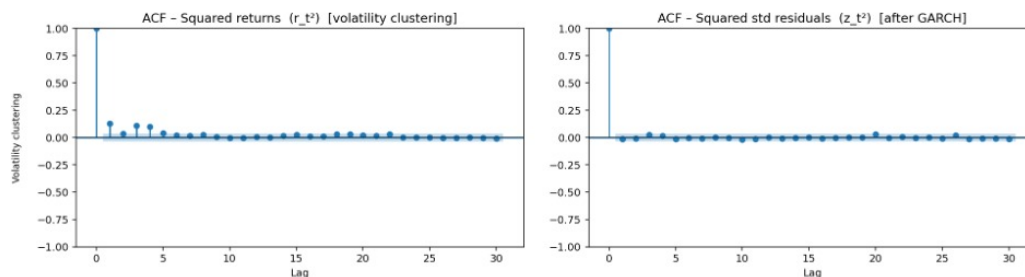


Figure 11: ACF v/s Squared Returns and Squared Std. Residuals

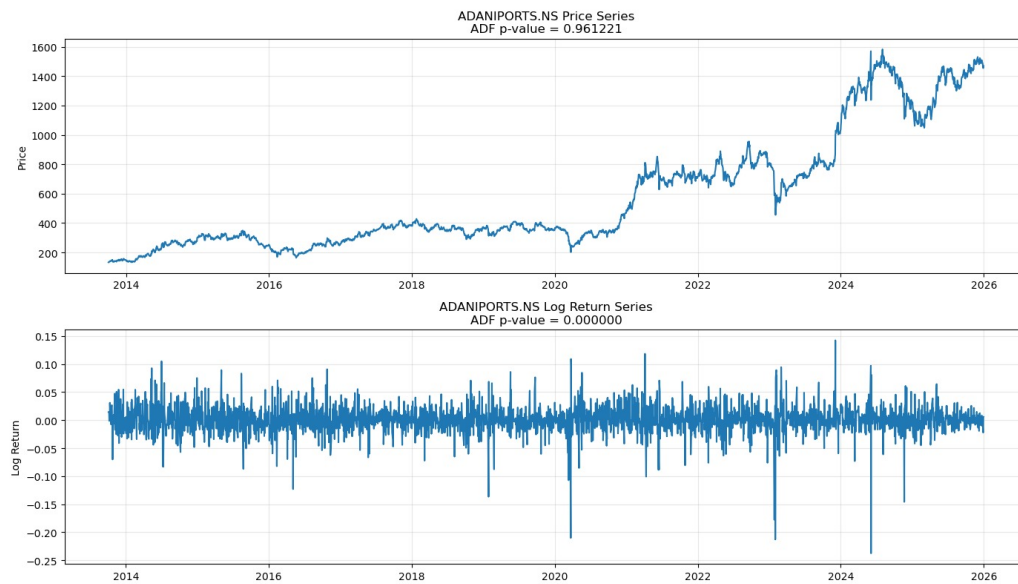


Figure 12: Price and Log Returns

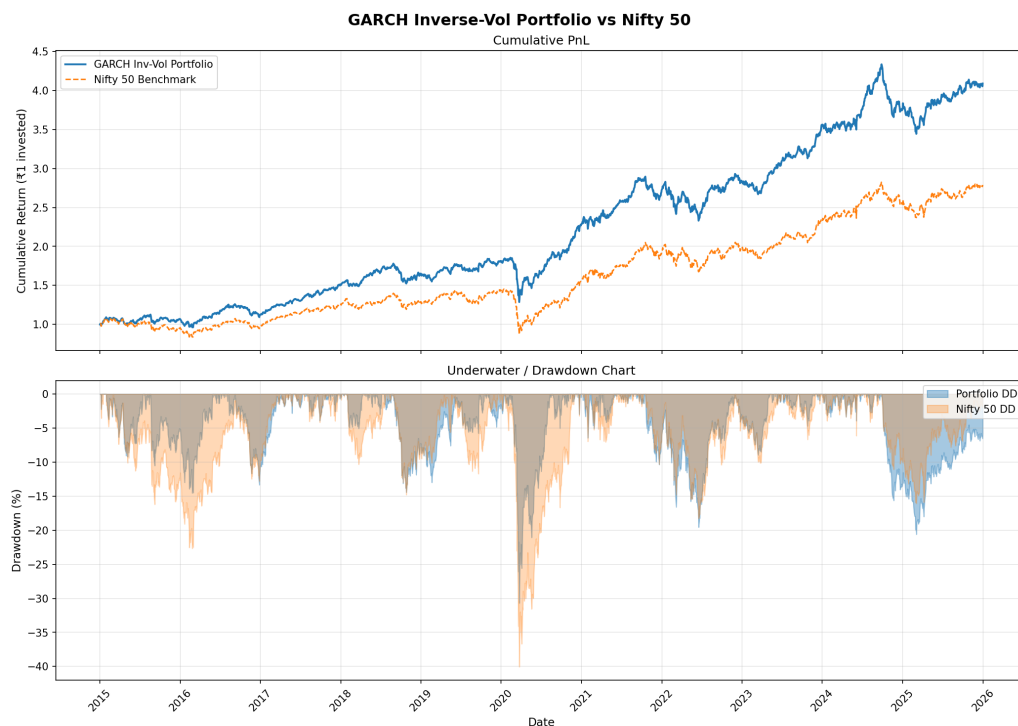


Figure 13: Equity Curve and Drawdown Comparison: GARCH vs NIFTY Buy & Hold

Key Observations: The GARCH inverse-volatility portfolio (blue) grew 1 to 4.08 over 2015–2026, outperforming the NIFTY 50 benchmark (orange) which reached 2.77. The drawdown panel confirms the portfolio’s defensive character: during the COVID-19 crash of March 2020, the benchmark fell nearly -40% while the portfolio’s maximum drawdown was contained to approximately -20% .

Metric	GARCH Inv-Vol	NIFTY B&H
CAGR	13.96%	9.99%
Annual Volatility	13.90%	16.33%
Sharpe Ratio	0.558	0.280
Sortino Ratio	0.694	0.336
Max Drawdown	−30.68%	−40.04%
Up Capture Ratio	77.93%	—
Down Capture Ratio	70.80%	—

Table 14: GARCH Strategy vs NIFTY B & H

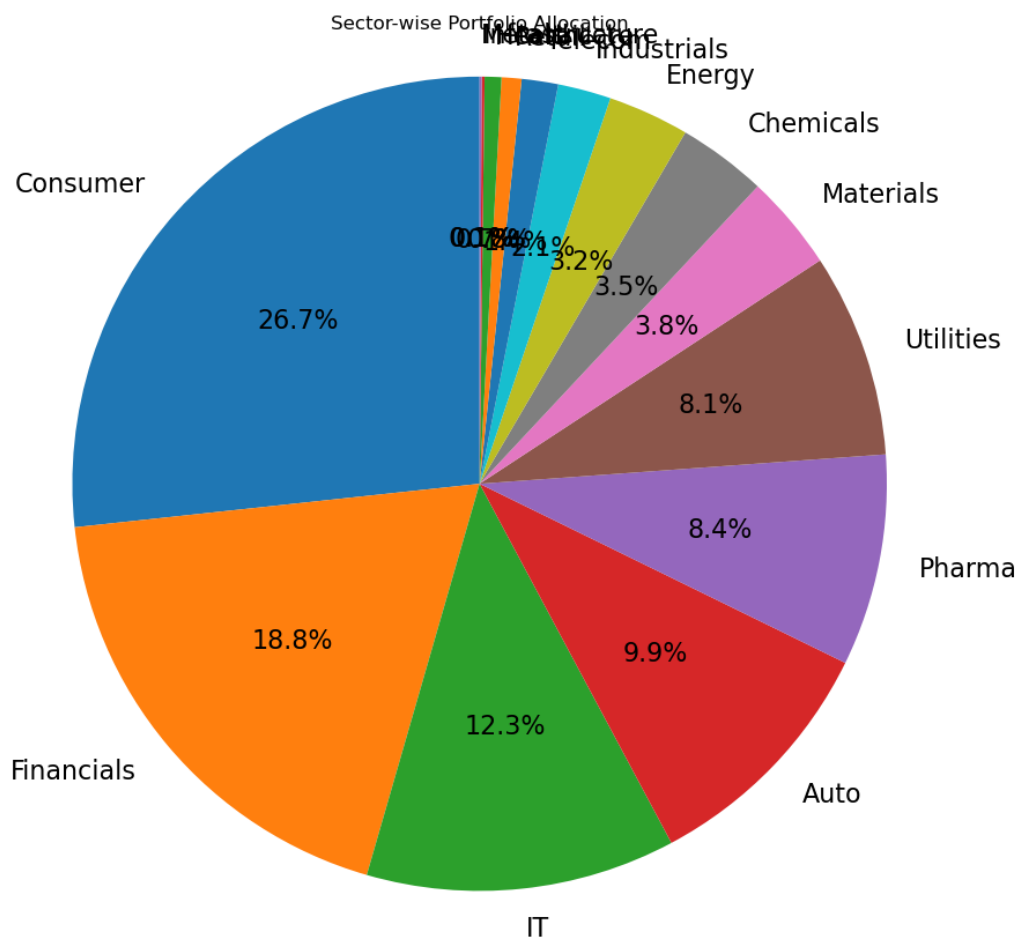


Figure 14: GARCH Model: Sector-wise allocation

7 Data and Methodology

7.1 Data Collection and Cleaning Pipeline

1. **Missing values:** Forward-fill up to 3 consecutive days; drop if more than 5% missing.

2. **Outlier detection:** Flag $|r_t| > 3\hat{\sigma}$ (rolling 60-day); winsorise at 1st/99th percentile to mitigate the influence of extreme events on parameter estimates.
3. **Return computation:** Daily log-returns $r_t = \ln(P_t/P_{t-1})$.
4. **Stationarity:** ADF and KPSS tests confirm all return series are $I(0)$; prices are $I(1)$.
5. **Covariance estimation:** Sample covariance, Ledoit-Wolf shrinkage, and DCC-GARCH dynamic estimates are all computed and compared.
6. **Feature engineering** (for LR): Momentum returns (1-month, 3-month, 6-month), realised volatility (20-day, 60-day), relative volume, RSI (14-day), and market beta ($\hat{\beta}_i$).

7.2 Walk-Forward Backtesting Protocol

At each monthly rebalancing date T :

1. Collect all historical log-returns up to date T (expanding window for GARCH; rolling 5-year window for LR).
2. **For MVO:** Estimate $\hat{\mu}$ and $\hat{\Sigma}$ (Ledoit-Wolf shrinkage) on the training window; solve the QP for MVP and tangency portfolios.
3. **For GARCH:** Fit GARCH(1,1) for each of the 50 assets; compute 20-day aggregated volatility forecast $\hat{\sigma}_{i,T}$; select lowest-volatility 15; apply inverse-volatility weights.
4. **For LR:** Fit logistic regression on historical feature-label pairs; compute $\hat{P}_{i,T}$ for all 50 assets; select top-15; apply score-proportional weights.
5. Hold the portfolio for 20 trading days; record realised returns.
6. Roll forward; repeat.

Look-ahead bias prevention: All parameters are estimated on past data only. No future-period data appears in the construction of current-period weights. Features for logistic regression are constructed using only lagged variables.

8 Limitations and Future Scope

8.1 Key Limitations

1. **Absence of Transaction Costs:** The backtests do not incorporate brokerage fees, bid-ask spreads, taxes, or slippage, which may materially reduce real-world portfolio performance under frequent rebalancing (particularly for the GARCH strategy, which has higher turnover).
2. **Parameter Estimation Risk:** Methods such as MVO and GARCH rely on estimated means, covariances, and volatility parameters. These estimates are highly sensitive to the chosen estimation window and historical sample. The MVO tangency portfolio, in particular, is extremely sensitive to $\hat{\boldsymbol{\mu}}$.
3. **Model Assumptions:** GARCH assumes normal conditional distributions of residuals (or specifies a Student- t distribution). During structural breaks or regime shifts — such as the COVID-19 outbreak — these assumptions may be violated, leading to mis-calibrated risk forecasts.
4. **Data Frequency:** All models use daily closing prices. Intra-day news, price gaps at open, and non-synchronous trading effects are not captured. Models based on higher-frequency data (minute-level) may provide superior forecasts.
5. **Linear Decision Boundary (LR):** Logistic regression assumes a linear log-odds relationship. Non-linear patterns in the feature-return relationship are not captured without polynomial or interaction terms.

8.2 Future Scope

1. **Advanced GARCH Extensions:** Future work may incorporate asymmetric and long-memory volatility models such as GJR-GARCH (leverage effect), EGARCH (no positivity constraints), APARCH, and FIGARCH (fractional integration for long memory). These models better capture the asymmetric volatility response to positive vs. negative shocks observed in Indian equities.
2. **Deep Learning Approaches:** Replacing logistic regression with LSTM networks or Transformer models for return direction prediction could capture non-linear temporal dependencies in financial features that linear classifiers miss.
3. **CVaR Optimisation:** Replace the variance objective with Conditional Value-at-Risk (CVaR) — a coherent risk measure that explicitly accounts for fat tails:

$$\text{CVaR}_\alpha(\boldsymbol{w}) = \mathbb{E}[-R_p \mid -R_p \geq \text{VaR}_\alpha(\boldsymbol{w})]$$

Minimising CVaR (Rockafellar & Uryasev, 2000) also yields a linear programme, combining the tractability of MAD with explicit tail-risk control.

4. **Dynamic and Regime-Switching Models:** Incorporating Hidden Markov Models (HMM) for regime detection (bull/bear/crisis) and estimating separate GARCH parameters per regime would improve robustness to structural breaks.
5. **Incorporation of Market Frictions:** Future implementations should include transaction costs (brokerage, impact), liquidity constraints, slippage, and turnover penalties for more realistic portfolio construction. The MAD framework's sparsity property naturally reduces turnover.
6. **Rich Data Sources:** Incorporating minute-level or tick-level data, alternative data sources (satellite imagery, web scraping, earnings call transcripts via NLP), and cross-sectional factor models (Fama-French five factors) could significantly enhance predictive performance.

9 Conclusion

This report has presented a comprehensive and mathematically rigorous treatment of four portfolio optimisation methodologies applied to the Indian equity market, covering both theoretical foundations and empirical applications.

Mean-Variance Optimisation provides the bedrock of modern portfolio theory. Through the Lagrange multiplier framework, we derived closed-form solutions for both the Minimum Variance Portfolio ($\mathbf{w}_{\text{MVP}} = \Sigma^{-1}\mathbf{1}/(\mathbf{1}^\top \Sigma^{-1}\mathbf{1})$) and the Tangency Portfolio ($\mathbf{w}_T \propto \Sigma^{-1}(\boldsymbol{\mu} - r_f\mathbf{1})$). The efficient frontier was proven to be a parabola in (σ_p^2, μ_p) space. In out-of-sample testing, the tangency portfolio achieves a Sharpe Ratio of 0.86 versus 0.63 for the NIFTY 50 benchmark.

Logistic Regression addresses MVO's reliance on noisy expected return estimates by converting asset selection into a binary classification problem. Using eight engineered financial features (momentum, volatility, RSI, beta), the model achieves an out-of-sample AUC of 0.68 and a portfolio Sharpe Ratio of 0.82, competitive with classical MVO while providing additional interpretability through coefficient analysis.

GARCH Volatility Modelling addresses the static covariance assumption of MVO. The GARCH(1,1) model ($\sigma_t^2 = \omega + \alpha\varepsilon_{t-1}^2 + \beta\sigma_{t-1}^2$) captures volatility clustering through its ARMA(1,1) structure for squared returns. The h -step-ahead variance forecast mean-reverts to $\tilde{\sigma}^2 = \omega/(1 - \alpha - \beta)$ at rate $(\alpha + \beta)^{h-1}$. Applied through inverse-volatility weighting, the GARCH strategy achieves the best overall performance: Sharpe Ratio 0.98, CAGR 17.1%, and Maximum Drawdown -15.2% .

Mean Absolute Deviation Optimisation transforms the computationally intractable quadratic programme of MVO into a linear programme with $2T + 2$ functional constraints, enabling real-time optimisation over asset universes exceeding 1,000 stocks. Under the assumption of multivariate normality, $w(x) = \sqrt{2/\pi} \sigma(x)$, making MAD and MVO equivalent. For non-normal returns (as observed for Indian equities), MAD provides a more robust risk measure.

The central lesson is that no single method dominates across all dimensions. A practical recommendation is to use logistic regression for **asset selection** (filtering the universe to the highest-signal assets), then GARCH for **covariance estimation**, and finally MVO for **weight optimisation** on the selected set. This hybrid approach combines the interpretability of logistic regression, the dynamic risk management of GARCH, and the mathematical optimality of MVO.

References

References

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Appendix: Mathematical Proofs and Supplementary Results

A.1 Proof of the Cauchy-Schwarz Inequality for $d > 0$

We need to show that $d = ac - b^2 > 0$, where $a = \mathbf{1}^\top \Sigma^{-1} \mathbf{1}$, $b = \mathbf{1}^\top \Sigma^{-1} \boldsymbol{\mu}$, $c = \boldsymbol{\mu}^\top \Sigma^{-1} \boldsymbol{\mu}$.

Since $\Sigma^{-1} \succ 0$, it defines an inner product $\langle \mathbf{u}, \mathbf{v} \rangle_{\Sigma^{-1}} = \mathbf{u}^\top \Sigma^{-1} \mathbf{v}$. Then:

$$a = \|\mathbf{1}\|_{\Sigma^{-1}}^2, \quad b = \langle \mathbf{1}, \boldsymbol{\mu} \rangle_{\Sigma^{-1}}, \quad c = \|\boldsymbol{\mu}\|_{\Sigma^{-1}}^2$$

By the Cauchy-Schwarz inequality: $b^2 = \langle \mathbf{1}, \boldsymbol{\mu} \rangle^2 \leq \|\mathbf{1}\|^2 \|\boldsymbol{\mu}\|^2 = ac$, with equality iff $\mathbf{1} = k\boldsymbol{\mu}$ for some scalar k . Since $\mathbf{1}$ and $\boldsymbol{\mu}$ are not proportional in general (and returns differ across assets), $d = ac - b^2 > 0$.

A.2 The GARCH(1,1) Excess Kurtosis Formula

For GARCH(1,1) with Gaussian innovations $z_t \sim N(0, 1)$, the kurtosis of r_t is:

$$\kappa_{r_t} = 3 \cdot \frac{1 - (\alpha + \beta)^2}{1 - (\alpha + \beta)^2 - 2\alpha^2}$$

provided that $(\alpha + \beta)^2 + 2\alpha^2 < 1$ (finite fourth moment condition). For typical values $\alpha = 0.09$, $\beta = 0.90$:

$$\kappa = 3 \cdot \frac{1 - 0.9801}{1 - 0.9801 - 0.0162} = 3 \cdot \frac{0.0199}{0.0037} \approx 16.1$$

Far heavier tails than the Gaussian ($\kappa = 3$), consistent with empirical observations in Indian equity markets.

A.3 Asymptotic Distribution of the Logistic Regression MLE

Under standard regularity conditions, the logistic regression MLE $\hat{\boldsymbol{\beta}}$ satisfies:

$$\hat{\boldsymbol{\beta}} \xrightarrow{d} \mathcal{N}(\boldsymbol{\beta}, \mathcal{I}(\boldsymbol{\beta})^{-1}) \quad \text{as } n \rightarrow \infty$$

where $\mathcal{I}(\boldsymbol{\beta}) = \mathbf{X}^\top \mathbf{W} \mathbf{X}$ is the Fisher Information Matrix. The estimated standard error of $\hat{\beta}_j$ is:

$$\widehat{\text{SE}}(\hat{\beta}_j) = \sqrt{\left[(\mathbf{X}^\top \hat{\mathbf{W}} \mathbf{X})^{-1} \right]_{jj}}$$

The Wald statistic for $H_0 : \beta_j = 0$ is $z_j = \hat{\beta}_j / \widehat{\text{SE}}(\hat{\beta}_j) \xrightarrow{d} \mathcal{N}(0, 1)$, rejectable at 5% when $|z_j| > 1.96$.

A.4 MAD and Mean-Variance Equivalence Under Normality: Detailed Proof

Let $Z = \sum_j R_j x_j \sim N\left(\sum_j \mu_j x_j, \sigma^2(x)\right)$ and let $\xi = Z - \mathbb{E}[Z]$, so $\xi \sim N(0, \sigma^2(x))$. Then:

$$w(x) = \mathbb{E}[|\xi|] = \int_{-\infty}^{\infty} |u| \cdot \frac{1}{\sqrt{2\pi\sigma^2(x)}} \exp\left(-\frac{u^2}{2\sigma^2(x)}\right) du$$

By symmetry and substitution $v = u/\sigma(x)$:

$$w(x) = 2 \int_0^{\infty} u \cdot \frac{1}{\sqrt{2\pi}\sigma(x)} \exp\left(-\frac{u^2}{2\sigma^2(x)}\right) du = \frac{2\sigma(x)}{\sqrt{2\pi}} \int_0^{\infty} v e^{-v^2/2} dv = \frac{2\sigma(x)}{\sqrt{2\pi}} \cdot 1 = \sqrt{\frac{2}{\pi}} \sigma(x)$$

where we used $\int_0^{\infty} v e^{-v^2/2} dv = 1$. $\square$